UNIT-II
WIRELESS SENSOR NETWORK

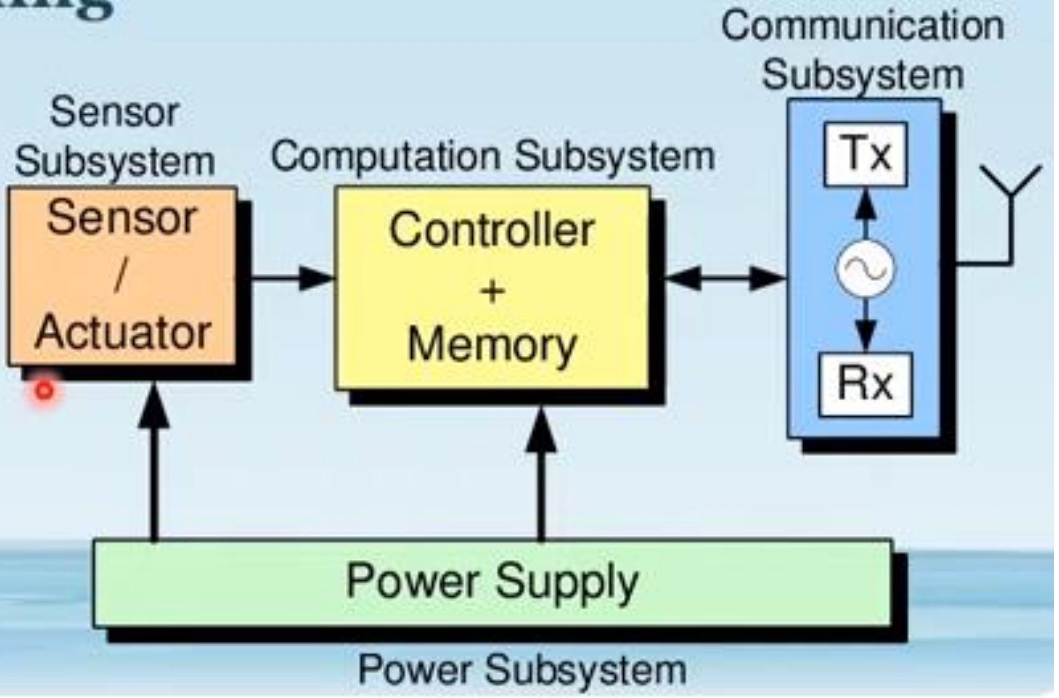
Transceiver Design Considerations in WSNs

Some crucial points influencing physical layer design in wireless sensor networks are:

1. Low power consumption.
2. First consequence: small transmit power and thus a small transmission range
3. Second consequence: Most hardware should be switched off or operated in a low-power standby mode most of the time.
4. Comparably low data rates, on the order of tens to hundreds kilobits per second, required.
5. Low implementation complexity and costs.
6. Low degree of mobility.

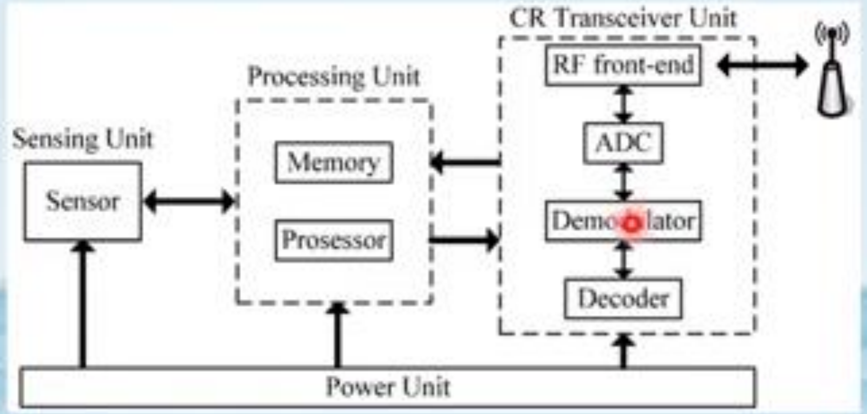
Transceiver design considerations in WSNs

1. Energy usage profile
2. Choice of modulation scheme
3. Dynamic modulation scaling
4. Antenna considerations



1.Energy Usage Profile

- The overall transceiver (RF front end and baseband part) consumes much more energy than is actually radiated;
- A transceiver working at frequencies beyond 1 GHz takes 10 to 100 mw of power to radiate 1 mw.
- For 2.4-GHz CMOS transceivers: for a radiated power of 0 dbm, the transmitter uses actually 32 mw, whereas the receiver uses even more, 38 mw.
- For the mica motes, 21 mw are consumed in transmit mode and 15 mw in receive mode



Cont.

- second key observation is that for small transmit powers the transmit and receive modes consume more or less the same power
- To reduce average power consumption in WSN, keeping the transceiver in idle mode all the time would consume significant amounts of energy.
- Therefore, it is important to put the transceiver into sleep state instead of just idling.
- there is the problem of the startup energy/startup time, which a transceiver has to spend upon waking up from sleep mode,
- For example, the μAMPS-1 transceiver needs a startup time of 466 ms and a power dissipation of 58 mW
- It depends on the traffic patterns and the behavior of the node MAC protocol to schedule the transceiver operational state properly.

Cont .

- **A third key observation is the relative costs of communications Vs computation in a sensor node.**
- **For the communication part on the BER requirements, range, transceiver type, and so forth,**
- **For the computation part on the processor type, the instruction mix, and so on.**
- **For the WIN nodes, 1500 to 2700 instructions can be executed per transmitted bit**
- **MEDUSA II nodes this ratio ranges from 220:1 up to 2900:1**
- **The bottom line is that computation is cheaper than communication!**

2.Choice of Modulation Scheme

- Several factors have to be balanced here: desirable data rate and symbol rate, the implementation complexity, the relationship between radiated power and target BER, and the expected channel characteristics.
- The higher the data rate offered by a transceiver, the smaller the time needed to transmit a given amount of data and, consequently, the smaller the energy consumption.
- A second important observation is that the power consumption of a modulation scheme depends much more on the symbol rate than on the data rate
- power consumption depends on the modulation scheme, with the faster Complementary Code Keying (CCK) modes consuming more energy than DBPSK and DQPSK
- Obviously, the desire for “high” data rates at “low” symbol rates calls for *m*-ary modulation schemes.

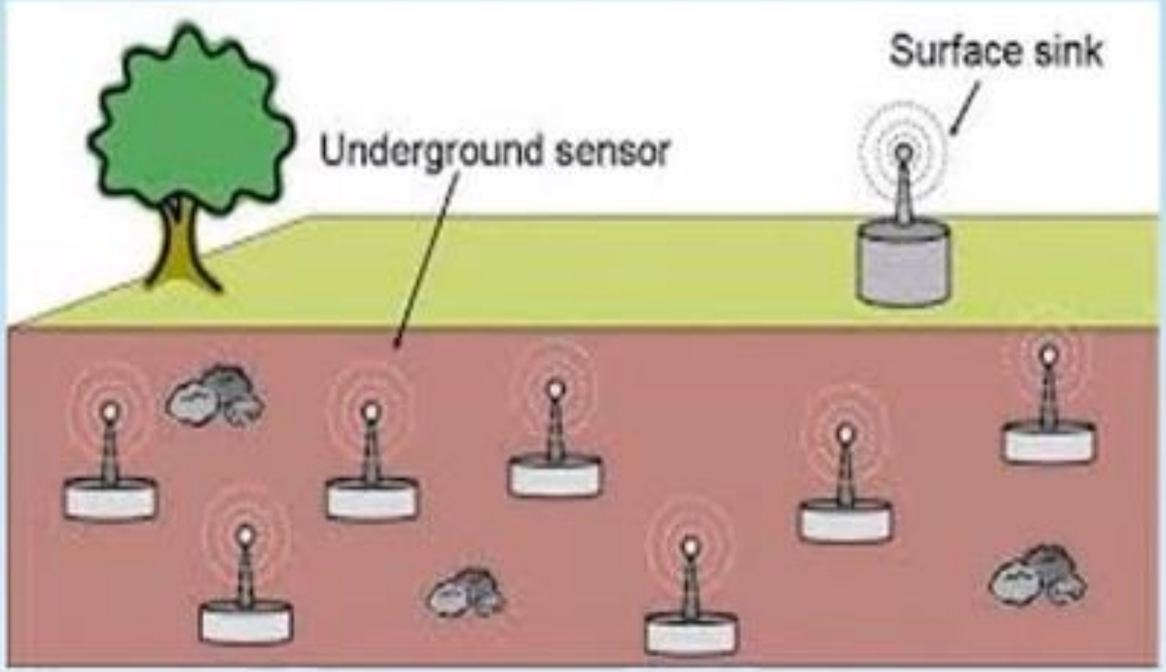
3.Dynamic Modulation Scaling

- it is interesting to consider methods to *adapt* the modulation scheme to the current situation.
- Such an approach, called dynamic modulation scaling,
- In particular, for the case of m-ary QAM and a target BER of 10^{-5}, a model has been developed that uses the symbol rate B and m no. of possible symbols.
- Clearly, the bit delay & energy per bit decreases for increasing B and m.
- The energy per bit depends much more on m than on B.
- both energy per bit and delay per bit are minimized for the maximum symbol rate B .

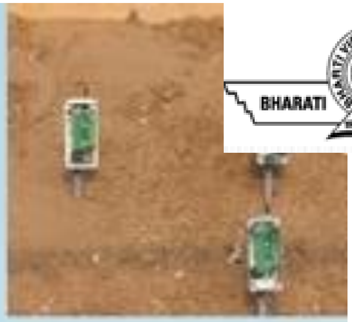
Truth Table

Binary input			8-QAM output	
Q	I	C	Amplitude	Phase
0	0	0	0.765 V	−135°
0	0	1	1.848 V	−135°
0	1	0	0.765 V	−45°
0	1	1	1.848 V	−45°
1	0	0	0.765 V	+135°
1	0	1	1.848 V	+135°
1	1	0	0.765 V	+45°
1	1	1	1.848 V	+45°

4.Antenna Considerations

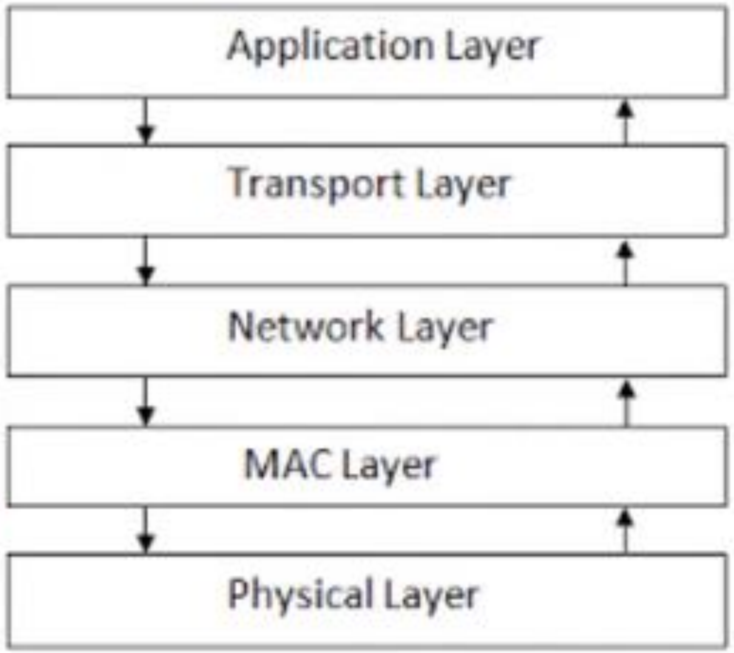


4.Antenna Considerations



- The desired **small size and the minimum number of antennas.**
- The antenna is much smaller than the carrier's wavelength, **it is hard to achieve good antenna efficiency.**
- With **unfriendly-sized antennas one must spend more transmit energy to obtain the same radiated energy.**
- Secondly, it will be **hard to place two antennas with suitable distance** to achieve receive diversity.
- For 2.4 GHz, this corresponds to a **spacing of between 5 to 6 cm between the antennas**, which is hard to achieve with smaller cases.
- In addition, **radio waves emitted from an antenna close to the ground** – typical in some applications – are faced with higher path-loss.
- Moreover, depending on the application, **antennas must not project from the casing of a node**, to avoid possible damage to it.
- These restrictions, in general, limit the achievable quality and characteristics of an antenna for wireless sensor nodes.

Medium Access Control (MAC) protocols



Importance of MAC Protocols

Medium access control (MAC) protocols:

MAC coordinate the times where a number of nodes access a shared communication medium.

MAC Protocols for Wireless Sensor Networks

- The single most important requirement **is energy efficiency**
- One important approach is to **switch the wireless transceiver into a sleep mode.**
- Specific requirements and design considerations for MAC protocols in wireless sensor networks.
- **Balance of requirements**
- **Major Sources of Energy Waste in MAC layer**
  1. **Collisions,**
  2. **Overhearing,**
  3. **Overhead,**
  4. **Idle listening.**

We discuss protocols addressing one or more of these issues.

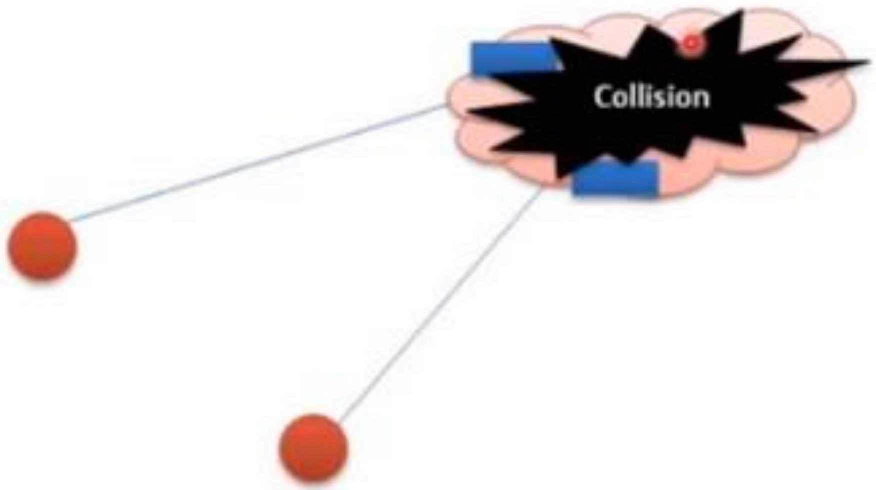
Cont.

- **Balance of requirements**
- The importance of energy efficiency for the design of MAC protocols is relatively new and many of the "classical" protocols like **ALOHA and CSMA** contain no provisions toward this goal.
- Other typical performance figures like **fairness, throughput, or delay tend to play a minor role in sensor networks.**
- Important requirements for MAC protocols are **scalability and robustness against frequent topology changes**

Energy problems on the MAC layer

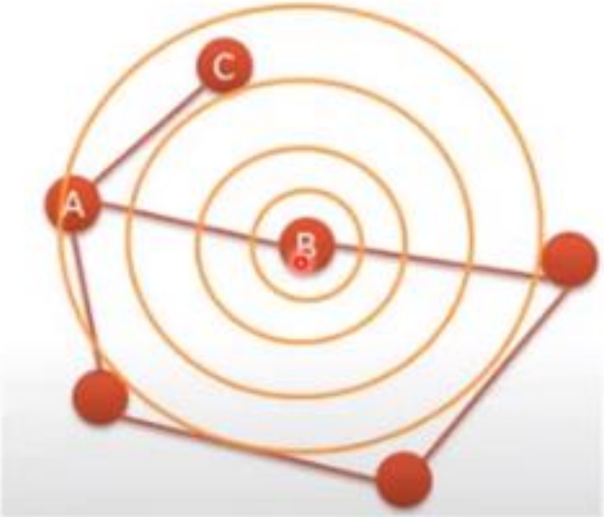
- A nodes transceiver consumes a significant share of energy.
- Transceiver can be in one of the four main states : **transmitting, receiving, idle and sleeping.**
- sleeping can be significantly cheaper than Idle state.
- MAC protocols objective is to reduce energy consumption.

a) Collision



Collision follow retransmission of packet which increase energy

b)Overhearing .



Node hear packet intended from other nodes

Cont.

c) Ideal listening

- Nodes listen to channel for possible traffic. If nothing is sensed then most of the time nodes is ideal.
- Ideal listening consume 50-100% energy required to receive packets.

d) Protocol overhead

- Protocol overhead is induced by MAC-related control frames like, RTS and CTS packets or request packets in demand assignment protocols, and furthermore by per-packet overhead like packet headers and trailers.

Types of MAC Protocols

- Low Duty Cycle Protocols And Wakeup Concepts –
  - S-MAC,
  - The Mediation Device Protocol
- Contention based protocols – PAMAS,
- Schedule based protocols – LEACH,
- IEEE 802.15.4 MAC protocol,

Issues in Designing MAC Protocol for Ad-hoc Wireless Networks:

➢ ***Bandwidth Efficiency:***

- Since the radio spectrum is limited, the bandwidth available for communication is also very limited. The MAC protocol must be designed in such a way that to maximize this bandwidth efficiency (the ratio of the bandwidth used for actual data transmission to the total available bandwidth).
- That is the uncommon bandwidth is utilized in an efficient manner.

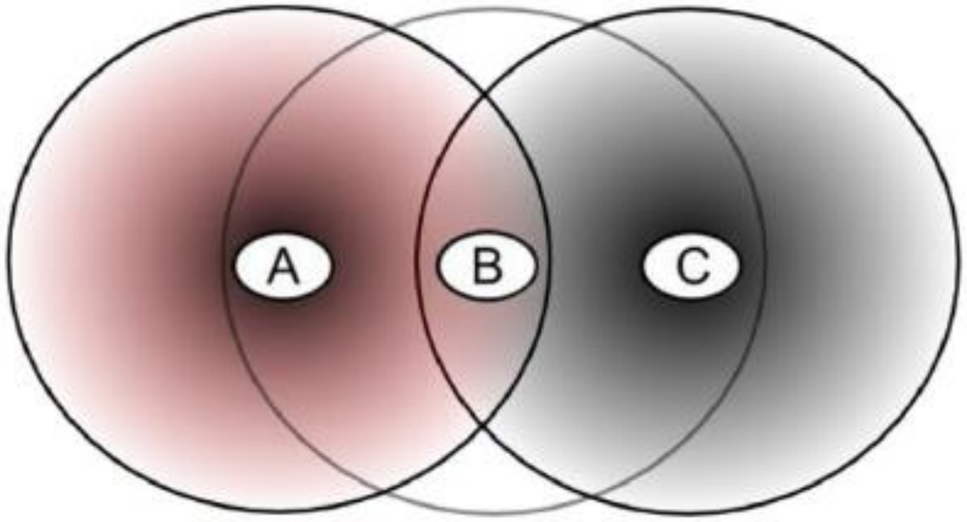
➢ ***Quality of Service Support(QoS):***

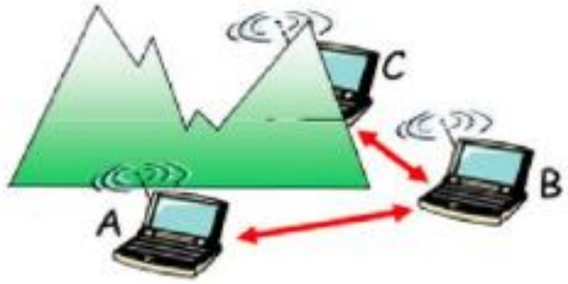
- Providing QoS support to data sessions in Ad-hoc networks is very difficult due to their characteristic nature of nodes mobility.
- Most of the time, Bandwidth reservation made at one point of time may become invalid once the node moves out of the region.
- The MAC protocol for Ad-hoc wireless networks that are to be used in such real-time applications must have resource reservation mechanism take care of nature of the wireless channel and the mobility of nodes.

Issues in Designing MAC Protocol for Ad-hoc Wireless Networks:

- ***Synchronization:***
  - The MAC protocol must take into consideration the synchronization between nodes in the network.
  - Synchronization is very important for bandwidth (time slot) reservations by nodes achieved by exchange of control packets.
- ***Hidden and Exposed Terminal Problems:***
  - The hidden terminal problem refers to the collision of packets at a receiving node due to the simultaneous transmission of those nodes.
  - The exposed terminal problem refers to the inability of a node, which is blocked due to transmission by a nearby transmitting node, to transmit to another node.
- ***Mobility of Nodes:***
  - This is a very important factor affecting the performance (throughput) of the protocol.
  - The MAC protocol obviously has no role to play in influencing the mobility of the nodes.

Hidden/Exposed terminal problem





Hidden terminal problem

- B, A hear each other
- B, C hear each other
- A, C can not hear each other

means A, C unaware of their interference at B

Issues in Designing MAC Protocol for Ad-hoc Wireless Networks:

➢ ***Error-Prone Shared Broadcast Channel:***

- Due to broadcast nature of the radio channel (transmissions made by a node are received by all nodes within its direct transmission range) there is a possibility of packet collisions is quite high in wireless networks.
- A MAC protocol should grant channel access to nodes in such a manner that collisions are minimized.

➢ ***Distributed Nature/Lack of Central Coordination***

- Ad hoc wireless networks do not have centralized coordinators because nodes keep moving continuously.
- Therefore, nodes must be scheduled in a distributed fashion for gaining access to the channel.
- The MAC protocol must make sure that the additional overhead, in terms of bandwidth consumption is not very high.

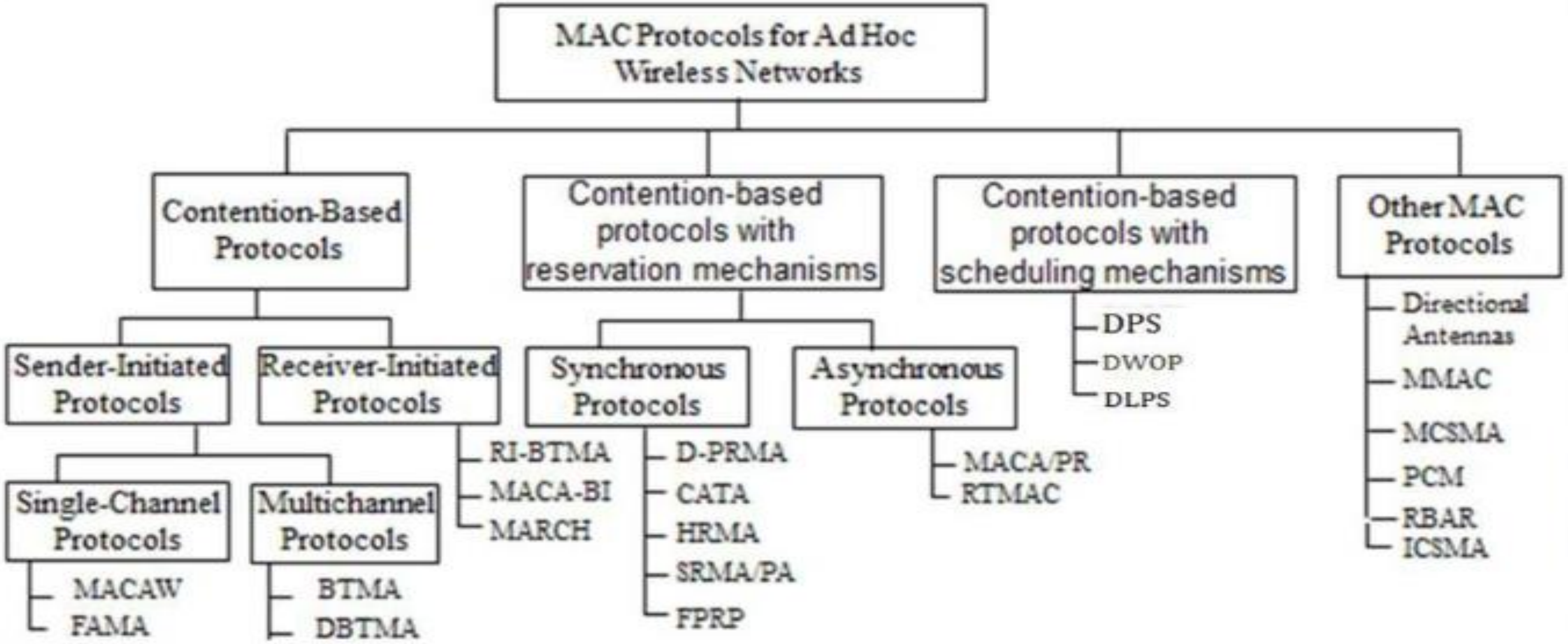
Design goals of a MAC Protocol for Ad-hoc Wireless Networks:

- The operation of the protocol should be distributed.
- The protocol should provide QoS support for real-time traffic.
- The access delay, which refers to the average delay experienced by any packet to get transmitted, must be kept low.
- The available bandwidth must be utilized efficiently.
- The protocol should ensure fair allocation of bandwidth to nodes.
- Control overhead must be kept as low as possible.
- The protocol should minimize the effects of hidden and exposed terminal problems.
- The protocol must be scalable to large networks.
- It should have power control mechanisms.
- The protocol should have mechanisms for adaptive data rate control.
- It should try to use directional antennas.
- The protocol should provide synchronization among nodes.

Classifications of MAC Protocols:

✓ Ad hoc network MAC protocols can be classified into three types:

- ❖ Contention-based protocols
- ❖ Contention-based protocols with reservation mechanisms
- ❖ Contention-based protocols with scheduling mechanisms
- ❖ Other MAC protocols



Classifications of MAC Protocols:

- ***Contention-based protocols:***
  - *Sender-initiated protocols:* Packet transmissions are initiated by the sender node.
    - *Single-channel sender-initiated protocols:* A node that wins the contention to the channel can make use of the entire bandwidth.
    - *Multichannel sender-initiated protocols*: The available bandwidth is divided into multiple channels.
  - *Receiver-initiated protocols:* The receiver node initiates the contention resolution protocol.

- ***Contention-based protocols with reservation mechanisms***
  - *Synchronous protocols*: All nodes need to be synchronized. Global time synchronization is difficult to achieve.
  - *Asynchronous protocols*: These protocols use relative time information for effecting reservations.

Classifications of MAC Protocols:

- ***Contention-based protocols with scheduling mechanisms:***
  - Node scheduling is done in a manner so that all nodes are treated fairly and no node is starved of bandwidth.
  - Scheduling-based schemes are also used for enforcing priorities among flows whose packets are queued at nodes.
  - Some scheduling schemes also consider battery characteristics.

- ***Other protocols are those MAC protocols :*** These are not strictly fall under the above categories.

Classifications of MAC Protocols:

➢ ***Contention-based protocols :***

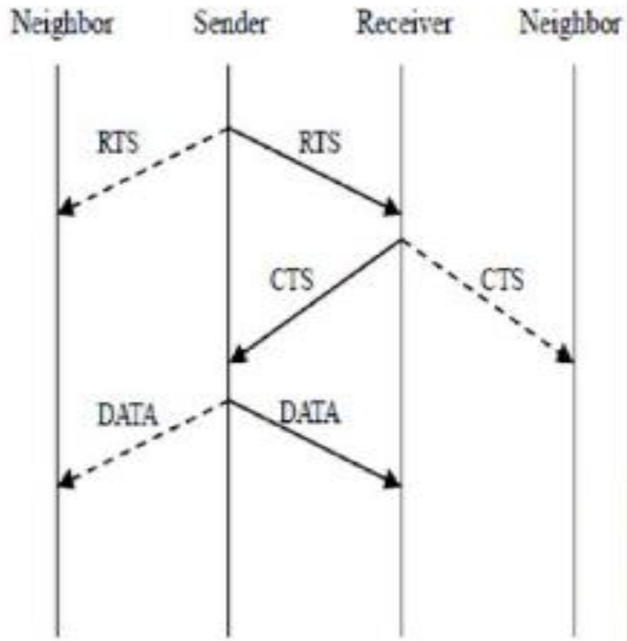
❖ ***Single-channel sender-initiated protocols:***

EXAMPLES: MACAW, FAMA

MACAW: A Media Access Protocol for Wireless LANs is based on MACA (Multiple Access Collision Avoidance) Protocol

MACA:-

- ✓ When a node wants to transmit a data packet, it first transmit a **RTS (Request To Send)** frame.
- ✓ The receiver node, on receiving the RTS packet, if it is ready to receive the data packet, transmits a **CTS (Clear to Send)** packet.
- ✓ Once the sender receives the CTS packet without any error, it starts transmitting the data packet.
- ✓ If a packet transmitted by a node is lost, the node uses the **Binary Exponential Back-off (BEB)** algorithm to back-off a random interval of time before retrying. The problem is solved by MACAW



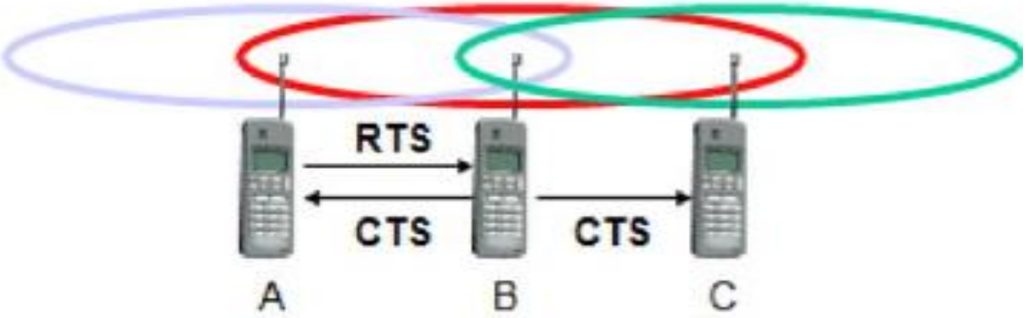
Classifications of MAC Protocols:

➢ ***Contention-based protocols :*** ***...cntd***

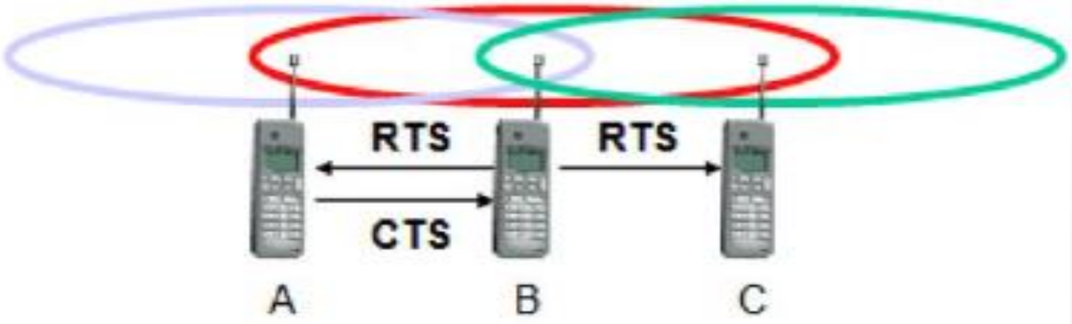
❖ ***Single-channel sender-initiated protocols:***

MACA EXAMPLES:

- MACA avoids the problem of hidden terminals
  - ✓ A and C want to send to B
  - ✓ A sends RTS first
  - ✓ C waits after receiving CTS from B



- MACA avoids the problem of exposed terminals
  - ✓ B wants to send to A, C to another terminal
  - ✓ now C does not have to wait for it cannot receive CTS from A



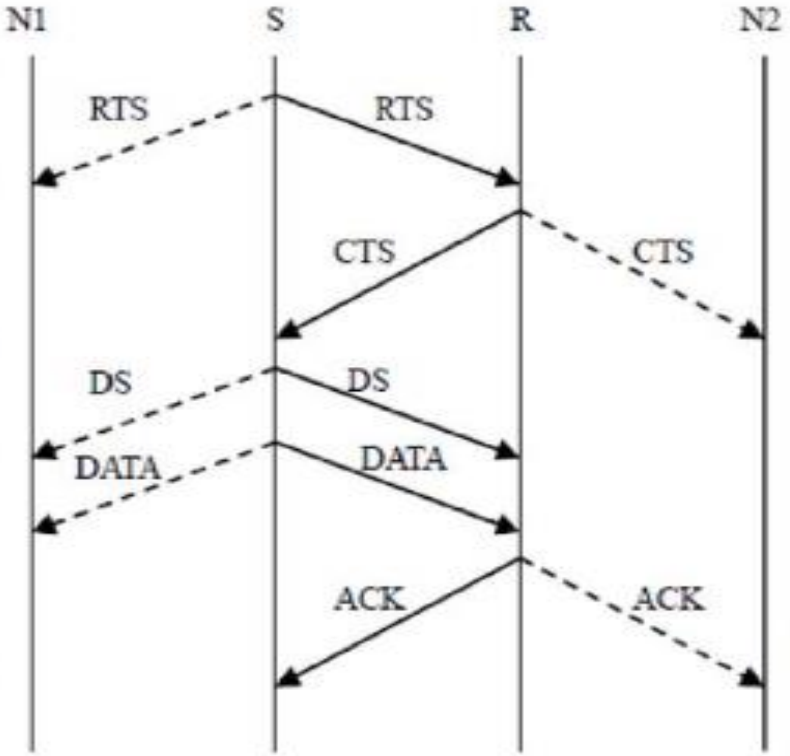
Classifications of MAC Protocols:

➢ ***Contention-based protocols :*** ***...cntd***

❖ ***Single-channel sender-initiated protocols:***

MACAW: (MACA for Wireless) is a revision of MACA.

- *The sender* transmits a **RTS (Request To Send)** frame if no nearby station transmits a RTS.
- *The receiver* replies with a **CTS (Clear To Send)** frame.
- *Neighbors*
  - see CTS, then keep quiet.
  - see RTS but not CTS, then keep quiet until the CTS is back to the sender.
- The receiver sends an ACK when receiving an frame.
  - Neighbors keep silent until see ACK.
- Collisions
  - There is no collision detection.
  - The senders know collision when they don't receive CTS.
  - They each wait for the exponential back-off time.



Classifications of MAC Protocols:

➢ ***Contention-based protocols :*** ***...cntd***

❖ ***Single-channel sender-initiated protocols:***

FAMA: Floor Acquisition Multiple Access Protocols.

- ✓ Channel access consists of a carrier-sensing operation and a collision avoidance
- ✓ Carrier-sensing by the sender, followed by the RTS-CTS control packet exchange.
  - Data transmission to be collision free, the duration of an RTS must be at least twice the maximum channel propagation delay
- ✓ Two FAMA protocol variants
  - RTS-CTS exchange with no carrier sensing (MACA)
  - RTS-CTS exchange with non-persistent carrier sensing (FAMA-NTR)

FAMA-NTR(Non-persistent Transmit Request)

- Before sending a packet, the sender senses the channel
- If channel is busy, the sender back-off a random time and retries later
- If the channel is free, the sender sends RTS and waits for a CTS packet
- If the sender cannot receive a CTS, it takes a random back-off and retries later
- If the sender receives a CTS, it can start transmission data packet
- In order to allow the sender to send a burst of packets, the receiver is made to wait a time duration τ seconds after a packet is received.

Classifications of MAC Protocols:

- ***Contention-based protocols :*** ***...cntd***

- ***Multi-channel sender-initiated protocols:***

- ***Busy Tone Multiple Access Protocols (BTMA):***
  - The transmission channel is split into two parts:
    - a data channel for data packet transmissions
    - a control channel used to transmit the busy tone signal
  - When a node is ready for transmission, it senses the channel to check whether the busy tone is active.
    - If not, it turns on the busy tone signal and starts data transmissions.
    - Otherwise, it reschedules the packet for transmission after some random rescheduling delay.

- ***Dual Busy Tone Multiple Access Protocol (DBTMAP)*** is an extension of the BTMA scheme.
  - a data channel for data packet transmissions
  - a control channel used for control packet transmissions (RTS and CTS packets) and also for transmitting the busy tones.
  - Use two busy tones on the control channel, BTt and BTr.
    - BTt : indicate that it is transmitting on the data channel
    - BTr: indicate that it is receiving on the data channel
  - Two busy tone signals are two sine waves at different frequencies

Classifications of MAC Protocols:

➢ ***Contention-based protocols :*** ...cntd

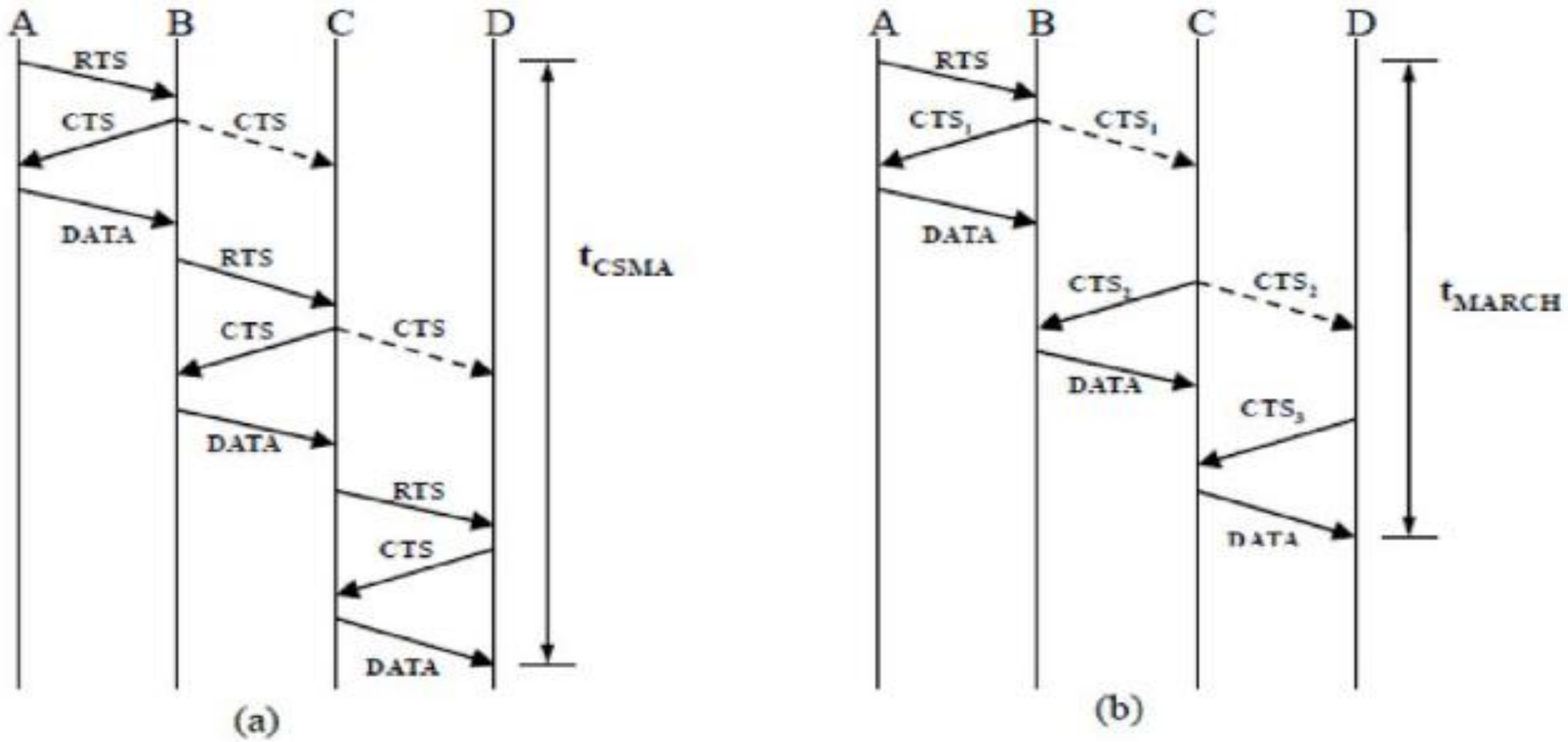
❖ ***Receiver-initiated protocols:***

- **RI-BTMA**: Receiver-Initiated Busy Tone Multiple Access Protocol
  - ✓ The transmission channel is split into two:
    - a data channel for data packet transmissions
    - a control channel used for transmitting the busy tone signal
  - ✓ A node can transmit on the data channel only if it finds the busy tone to be absent on the control channel.
  - ✓ The data packet is divided into two portions: a preamble and the actual data packet.
- **MACA-BI:** MACA-By Invitation
  - ✓ By eliminating the need for the RTS packet it reduces the number of control packets used in the MACA protocol which uses the three-way handshake mechanism.
- **MARCH:** Media Access with Reduced Handshake.

Classifications of MAC Protocols:

➢ ***Contention-based protocols :*** *...cntd*

❖ ***Receiver-initiated protocols:***



Handshake mechanism in (a) MACA and (b) MARCH

Classifications of MAC Protocols:

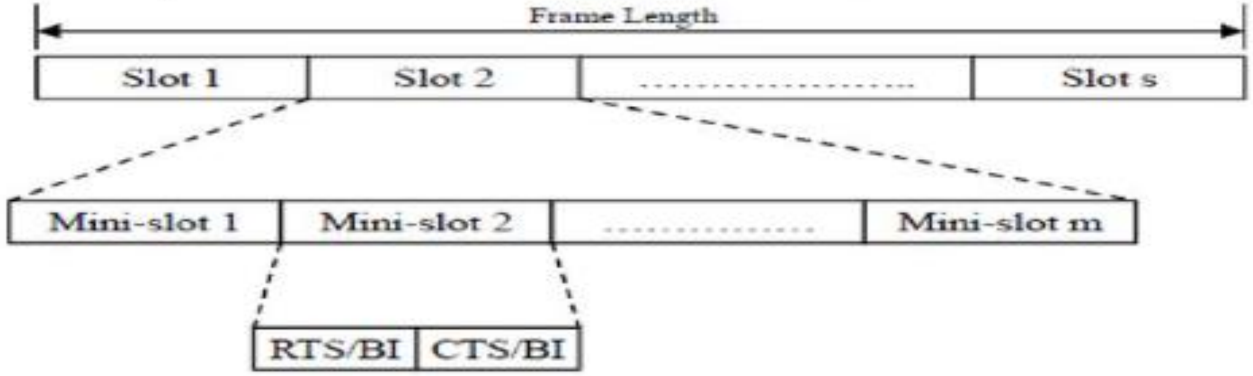
➢ ***Contention-based Protocols with Reservation Mechanism:***

- ✓ Contention occurs during the resource (bandwidth) reservation phase.
- ✓ Once the bandwidth is reserved, the node gets exclusive access to the reserved bandwidth.
- ✓ QoS support can be provided for real-time traffic.

❖ ***Synchronous protocols:***

❑ **Distributed Packet Reservation Multiple Access Protocol(D-PRMA)**

- It extends the centralized packet reservation multiple access (PRMA) scheme into a distributed scheme that can be used in ad hoc wireless networks.
- PRMA was designed in a wireless LAN with a base station.
- D-PRMA is a TDMA-based scheme. The channel is divided into fixed- and equal-sized frames along the time axis.



Classifications of MAC Protocols: ...cntd

➢ ***Contention-based Protocols with Reservation Mechanism:***

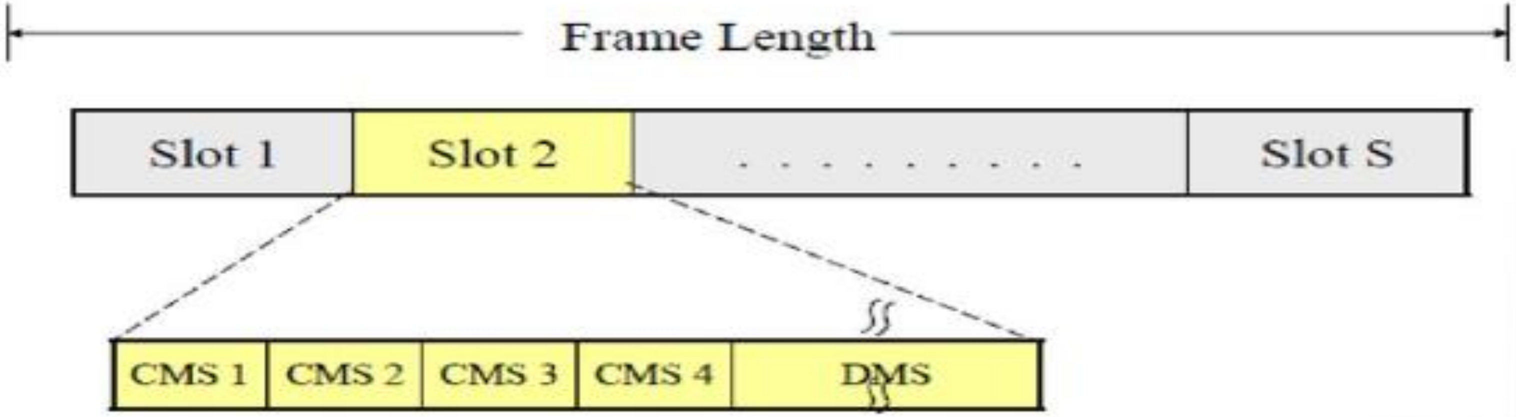
❖ ***Synchronous protocols:***

❑ **Collision Avoidance Time Allocation Protocol(CATA):**

✓ Support broadcast, unicast, and multicast transmissions simultaneously.

✓ Each frame consists of S slots and each slot is further divided into five Control Mini-Slots

- CMS1: Slot Reservation (SR)
- CMS2: RTS
- CMS3: CTS
- CMS4: Not To Send (NTS)
- DMS: Data transmission



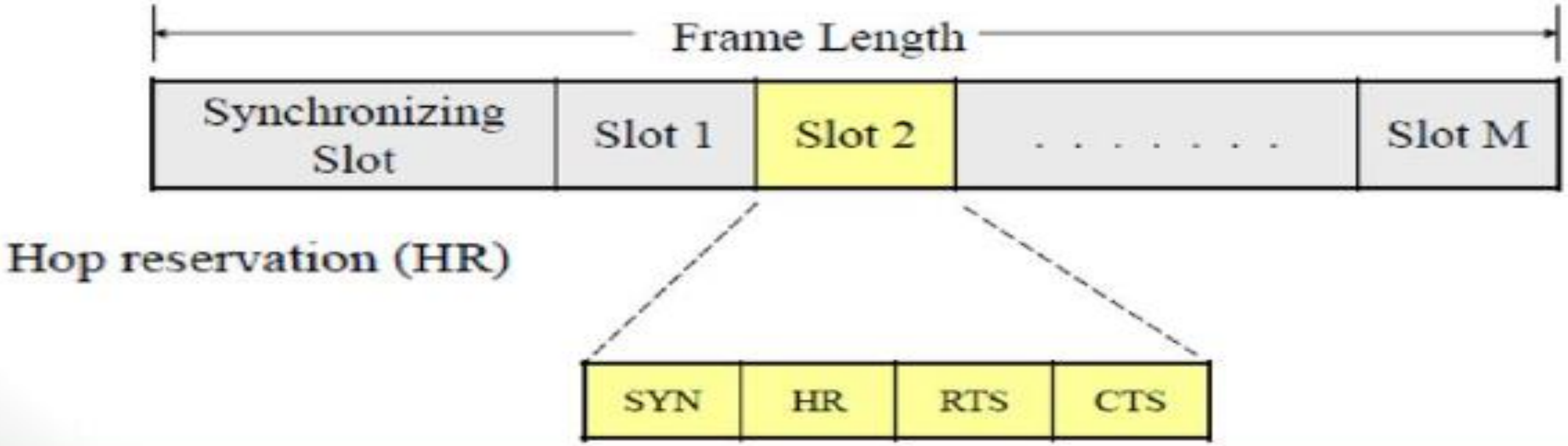
Classifications of MAC Protocols: ...cntd

➢ ***Contention-based Protocols with Reservation Mechanism:***

❖ ***Synchronous protocols:***

❑ Hop Reservation Multiple Access Protocol (HRMA):

- ✓ A multichannel MAC protocol which is based on half-duplex, very slow frequency-hopping spread spectrum (FHSS) radios
- ✓ Uses a reservation and handshake mechanism to enable a pair of communicating nodes to reserve a frequency hop, thereby guaranteeing collision-free data transmission.
- ✓ Can be viewed as a time slot reservation protocol where each time slot is assigned a separate frequency channel.



Classifications of MAC Protocols: ...cntd

➢ ***Contention-based Protocols with Reservation Mechanism:***

❖ ***Synchronous protocols:***

- **Soft Reservation Multiple Access with Priority Assignment (SRMA/PA):**
  - ✓ Developed with the main objective of supporting integrated services of real-time and non-real-time application in Ad-hoc networks.
  - ✓ Nodes use a collision-avoidance handshake mechanism and a soft reservation mechanism.
- **Five-Phase Reservation Protocol (FPRP)**
  - ✓ A single-channel TDMA based broadcast scheduling protocol.
  - ✓ Nodes uses a contention mechanism in order to acquire time slots.
  - ✓ The protocol assumes the availability of global time at all nodes.
  - ✓ The reservation takes five phases:
    - Reservation,
    - Collision Report,
    - Reservation Confirmation,
    - Reservation Acknowledgement,
    - Packing And Elimination Phase.

Classifications of MAC Protocols: ...cntd

➢ ***Contention-based Protocols with Reservation Mechanism:***

❖ ***Synchronous protocols:***

❑ **Five-Phase Reservation Protocol (FPRP)**

Five-phase protocol:

- **Reservation request**: send reservation request (RR) packet to dest.
- **Collision report**: if a collision is detected by any node, that node broadcasts a CR packet
- **Reservation confirmation**: a source node won the contention will send a RC packet to destination node if it does not receive any CR message in the previous phase
- **Reservation acknowledgment**: destination node acknowledge reception of RC by sending back RA message to source
- **Packing and elimination**: use packing packet and elimination packet.

Classifications of MAC Protocols: ...cntd

- ***Contention-based Protocols with Reservation Mechanism:***

❖ ***Asynchronous protocols:***

- **MACA with Piggy-Backed Reservation (MACA/PR):**
  - ✓ Provide real-time traffic support in multi-hop wireless networks
  - ✓ Based on the MACAW protocol with non-persistent CSMA
  - ✓ The main components of MACA/PR are:
    - A MAC protocol
    - A reservation protocol
    - A QoS routing protocol

- **Real-Time Medium Access Control Protocol (RTMAC)**
  - ✓ Provides a bandwidth reservation mechanism for supporting real-time traffic in ad-hoc wireless networks
  - ✓ RTMAC has two components
    - A MAC layer protocol is a real-time extension of the IEEE 802.11 DCF.
      - A medium-access protocol for best-effort traffic
      - A reservation protocol for real-time traffic
    - A QoS routing protocol is responsible for end-to-end reservation and release of bandwidth resources.

Classifications of MAC Protocols:

➢ ***Contention-based protocols with Scheduling Mechanism:***

✓ Protocols in this category focus on packet scheduling at the nodes and transmission scheduling of the nodes.

✓ The factors that affects scheduling decisions

- Delay targets of packets
- Traffic load at nodes
- Battery power

✓ Distributed priority scheduling and medium access in Ad Hoc Networks present two mechanisms for providing quality of service (QoS)

- **Distributed priority scheduling (DPS)** – Piggy-backs the priority tag of a node's current and head-of-line packets to the control and data packets
- **Multi-hop coordination** – Extends the DPS scheme to carry out scheduling over multi-hop paths.

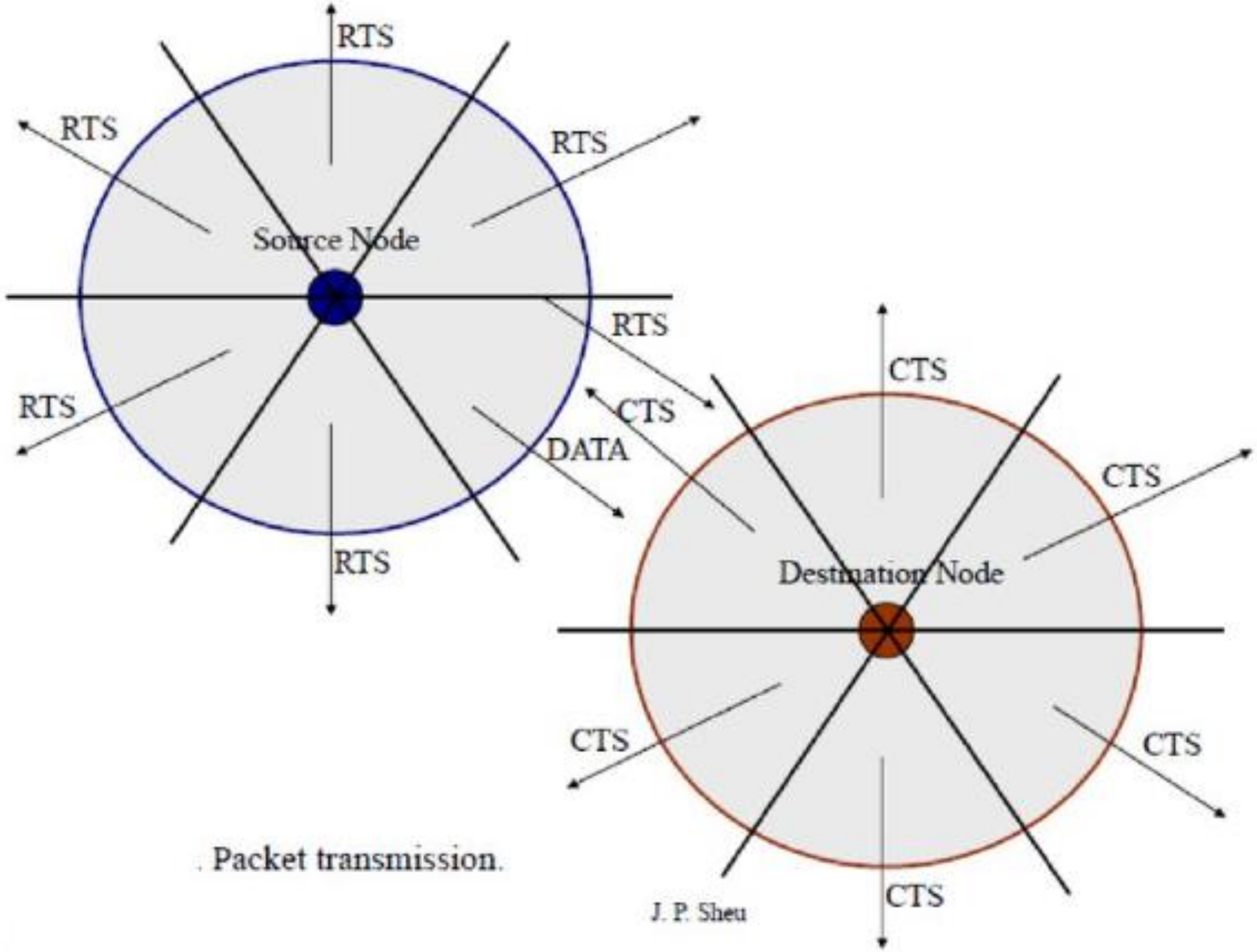
Classifications of MAC Protocols: ...cntd

➢ ***Contention-based protocols with Scheduling Mechanism:***

- **Distributed Wireless Ordering Protocol (DWOP)**
  - A media access scheme along with a scheduling mechanism based on the distributed priority scheduling scheme
- **Distributed Laxity-based Priority Scheduling (DLPS) Scheme**
  - Scheduling decisions are made based on the states of neighboring nodes and feed back from destination nodes regarding packet losses
  - Packets are recorded based on their uniform laxity budgets (ULBs) and the packet delivery ratios of the flows. The laxity of a packet is the time remaining before its deadline.

Classifications of MAC Protocols: ...cntd

➢ ***MAC Protocols that use directional Antennas:***



. Packet transmission.

Classifications of MAC Protocols: ...cntd

- ***Other MAC Protocols:***
  - ✓ **Multi-channel MAC Protocol (MMAC)**
    - Multiple channels for data transmission
    - There is no dedicated control channel.
    - Based on channel usage channels can be classified into three types: high, medium and low preference channels.
  - ✓ **Multi-channel Carrier Sense Multiple Access(MCSMA) MAC Protocol :**
    - The available bandwidth is divided into several channels
  - ✓ **Power Control MAC Protocol (PCM) for Ad Hoc Networks**
    - Allows nodes to vary their transmission power levels on a per-packet basis
  - ✓ **Receiver-based Autorate Protocol (RBAR)**
    - Use a rate adaptation approach
  - ✓ **Interleaved Carrier-Sense Multiple Access Protocol (ICSMA)**
    - The available bandwidth is split into tow equal channels
    - The handshaking process is interleaved between the two channels.

Note: *A directional antenna or beam antenna is an antenna which radiates or receives greater power in specific directions allowing for increased performance and reduced interference from unwanted sources.*

Note: *Omnidirectional refers to the notion(feeling) of existing in every direction. Omnidirectional antenna is that radiates equally in all directions.*

Note: *Handshaking is the exchange of information between two modems and the resulting agreement about which protocol to use that precedes each telephone connection.*

LOCATION DISCOVERY

- The location information of sensors has to be considered during aggregation of sensed data.

- This implies each node should know its location and couple its location information with the data in the messages it sends.

- A low-power , inexpensive, and reasonably accurate mechanism is needed for location discovery.

- A global positioning system (GPS) is not always feasible because it cannot reach nodes in dense foliage or indoors.

- It also consumes high power and makes sensor nodes bulkier.

- Two basic mechanisms of location discovery are :

1 . Indoor Localization :

- Indoor localization techniques use a fixed infrastructure to estimate the location of sensor nodes.

- Fixed beacon nodes are strategically placed in the field of observation, typically indoors, such as within a building.

- The randomly distributed sensors receive beacon signals from the beacon nodes and measure the signal strength, angle of arrival, and time difference between the arrival of different beacon signals.

- Using the measurements from multiple beacons, the nodes estimate their location. Some approaches use simple triangulation methods, while others require *a priori* database creation of signal measurements.

- The nodes estimate distances by looking up the database instead of performing computations. However, storage of the database may not be possible in each node, so only the BS may carry the database.

2 . Sensor Network Localization :

- In situations where there is no fixed infrastructure available and prior measurements are not possible, some of the sensor nodes themselves act as beacons.

- They have their location information, using GPS, and these send periodic beacons to other nodes.

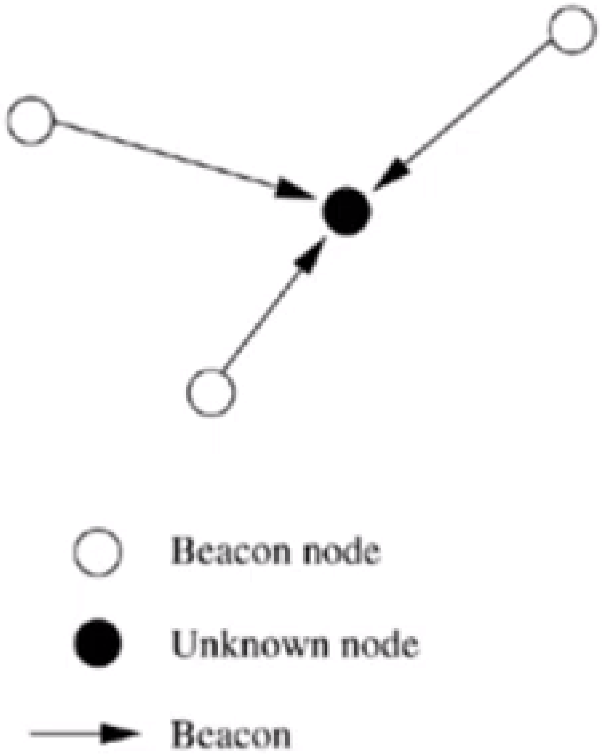
- Localization algorithms require techniques for location estimation depending on the beacon nodes location.

- These are called **multi-lateration (ML) techniques.**

- Some simple ML techniques are :

(i) Atomic ML:

- If a node receives three beacons, it can determine its position by a mechanism similar to GPS.



(ii) Iterative ML:

- Some nodes may not be in the direct range of three beacons.
- Once a node estimates its location, it sends out a beacon, which enables some other nodes to now receive at least three beacons.
- Iteratively, all nodes in the network can estimate their location.
- The drawback of this multi-hop method is that errors are propagated, hence estimation of location may not be accurate.

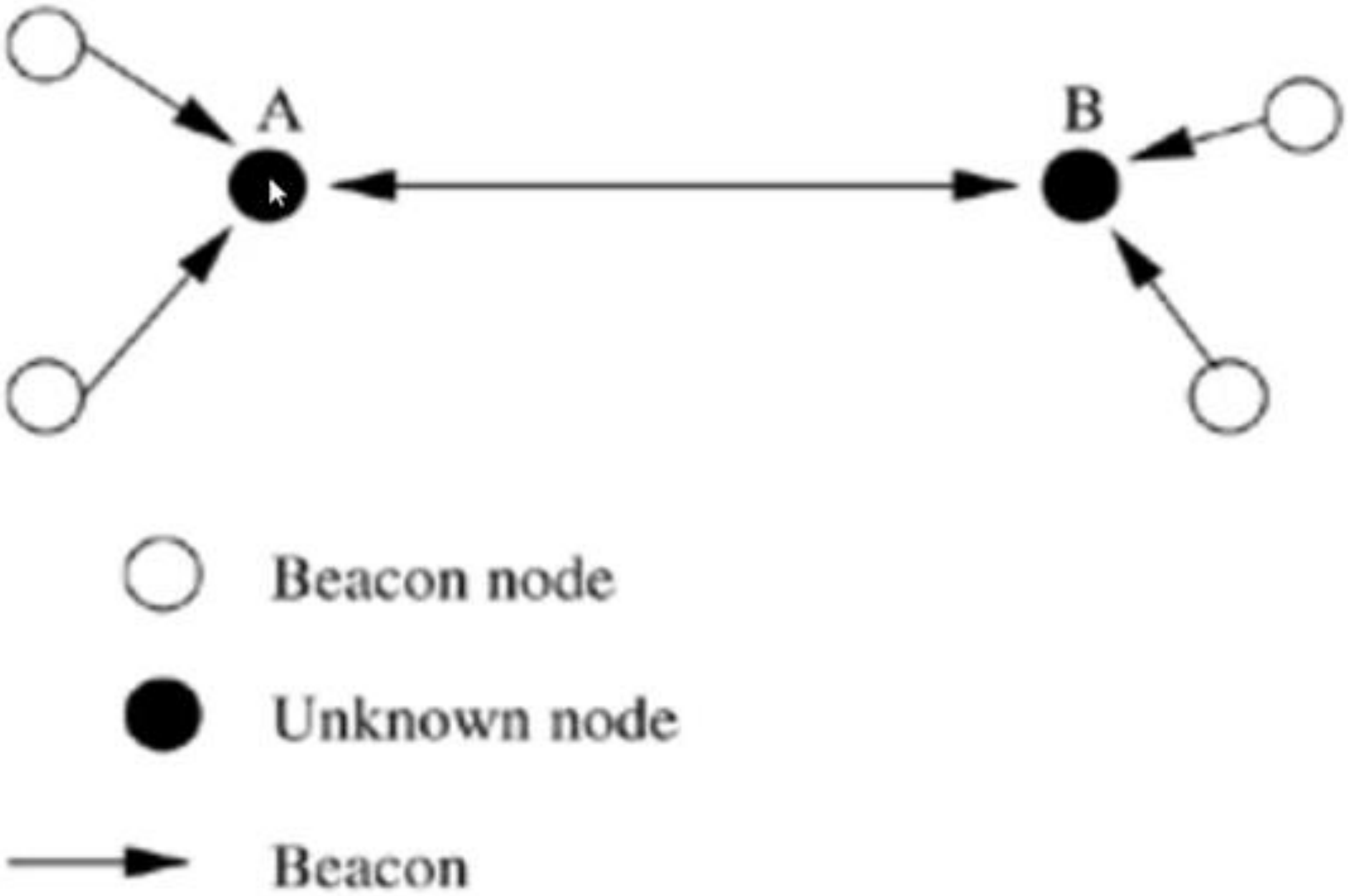
(iii) Collaborative ML:

- When two or more nodes cannot receive at least three beacons each, they collaborate with each other.

- In figure node A and node B have three neighbors each. Of the six participating nodes, four are beacons, whose positions are known.

- Hence, by solving a set of simultaneous quadratic equations, the positions of A and B can be determined.

Collaborative multi-lateration



S-MAC (Sensor- Medium Access Control)

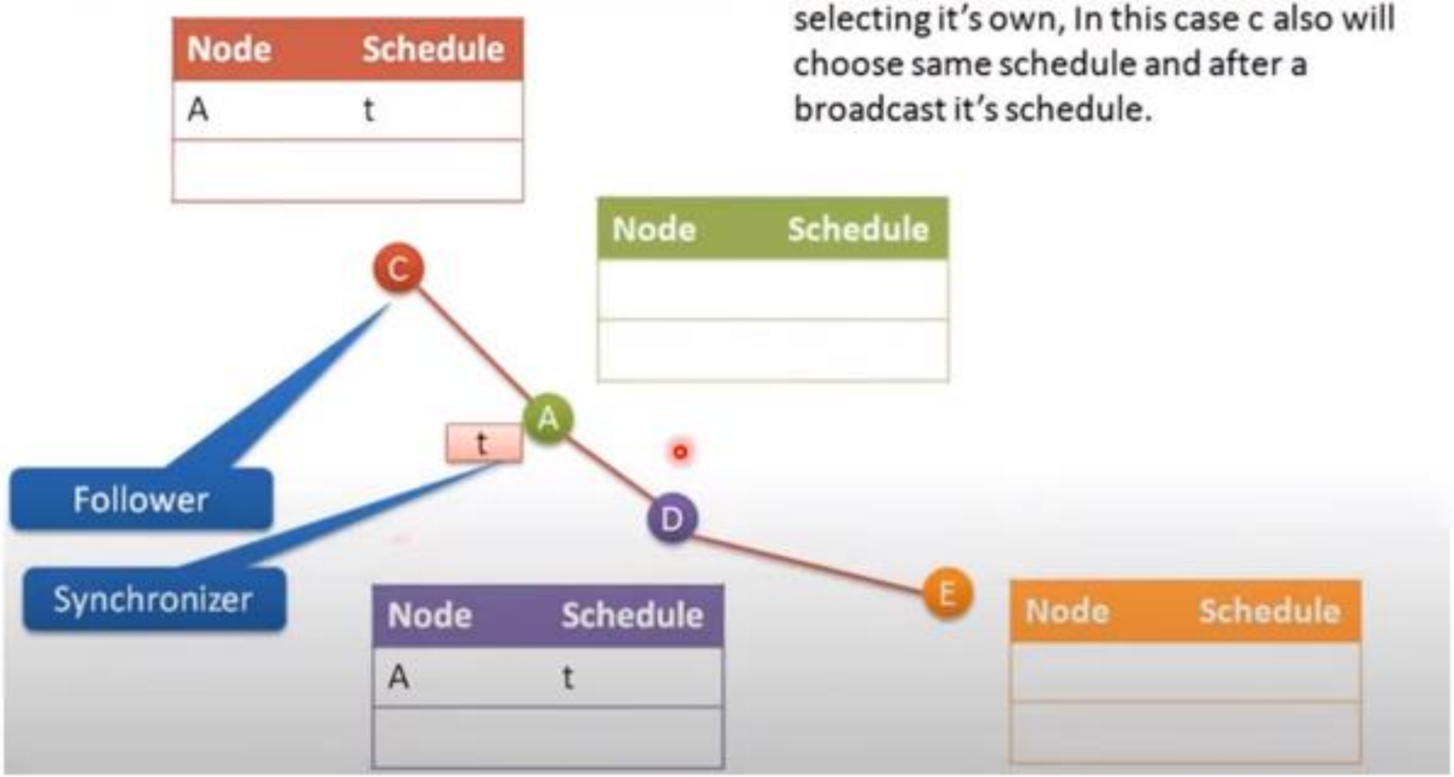
1. Periodic listen and sleep
2. Collision and overhearing avoidance
3. Message passing

<u>Periodic listen and sleep</u>

- Every node sleep for some time and then walkup to see any other node wants to talk to him
- During sleep time node **turn off its radio to save Energy**

Schedule maintenance

3. Node C received schedule before selecting it's own, In this case c also will choose same schedule and after a broadcast it's schedule.



Maintaining Synchronization

- **Collision avoidance?**

- Schedules are periodically broadcasted to neighbor
- Schedules are sent in SYNC packet.
- Listen time is divided into two potions one for SYNC and another for Data.

Maintaining Synchronization

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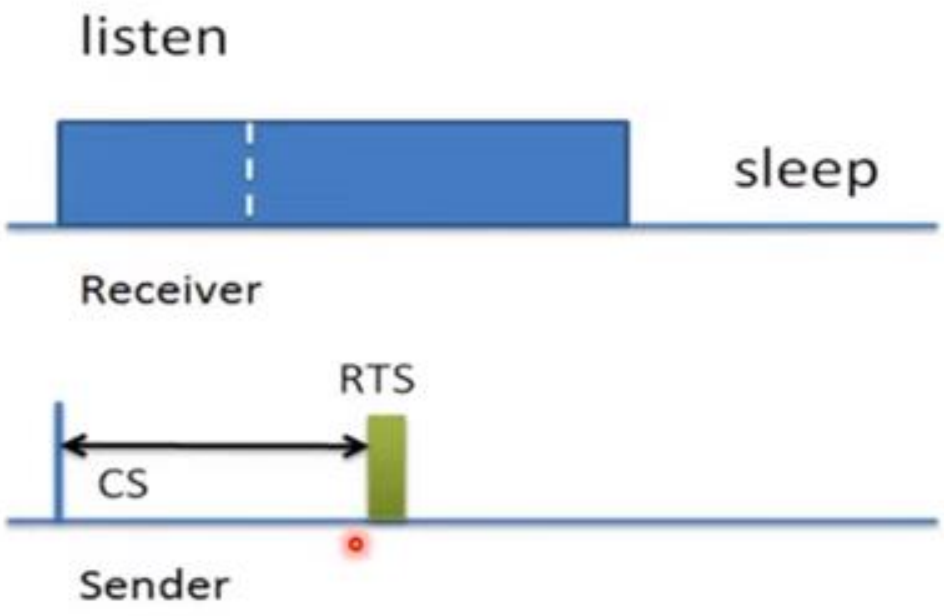
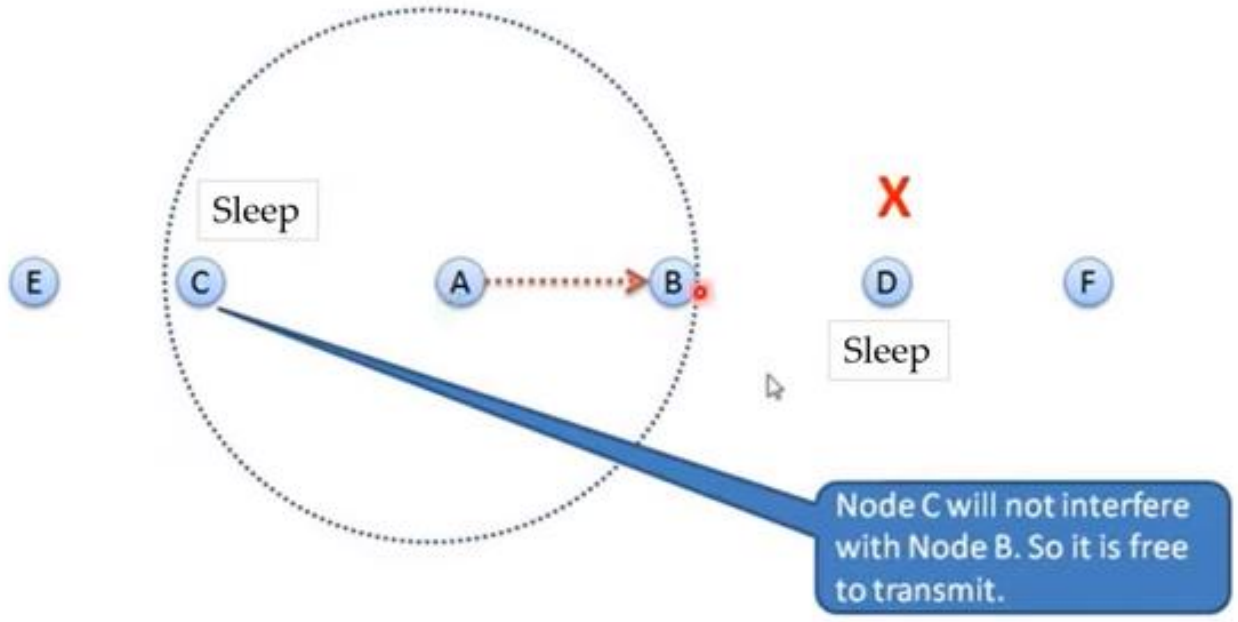
data sending



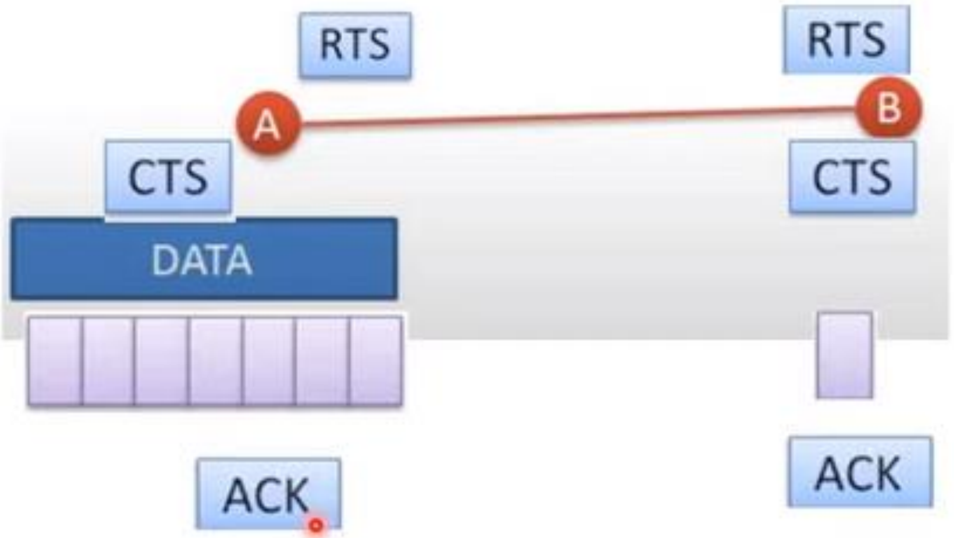
1. Time portion further divided in to slots.
2. Sender node start carrier sense when receiver start listening
3. Sender will select one mini slot randomly to finish CS process
4. Till that slot sender will sense that channel and if channel is free till that slot in next slot sender sends RTS packet

Overhearing Avoidance

All immediate neighbor of sender and receiver must sleep once they receive RTS/CTS packets



Message passing



RTS and CTS mechanism is used to send data

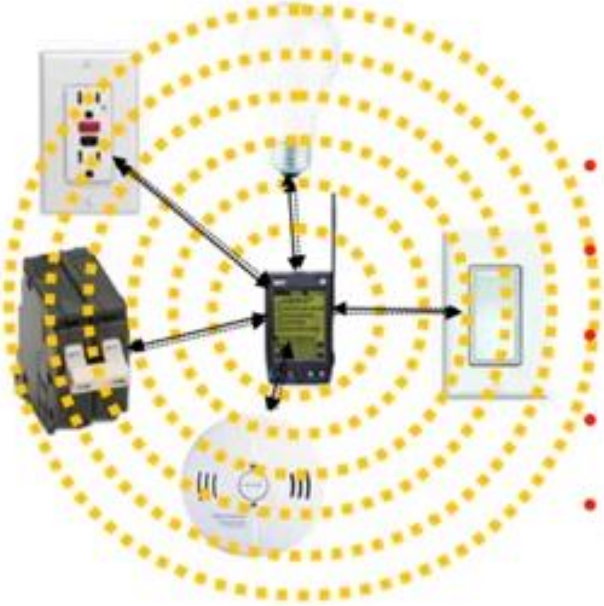
Types of MAC Protocols

- Low Duty Cycle Protocols And Wakeup Concepts –
  - S-MAC,
  - The Mediation Device Protocol
- Contention based protocols – PAMAS,
- Schedule based protocols – LEACH,
- IEEE 802.15.4 MAC protocol,

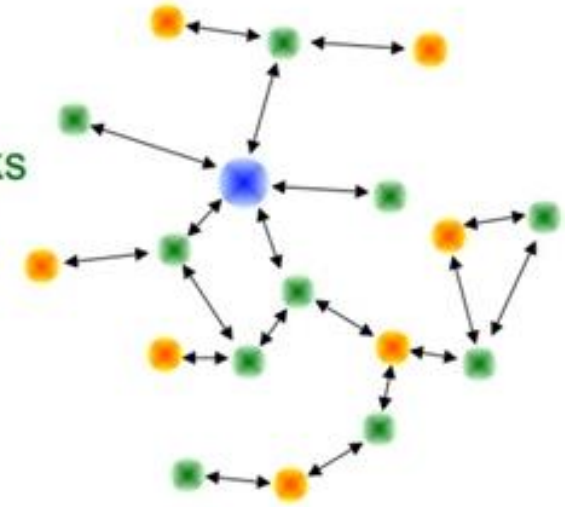
802.15.4 General Characteristics

- The standard covers the physical layer and the MAC layer of a **low-Data rate Wireless Personal Area Network (WPAN).**
- Star or Peer-to-Peer operation.
- Support for low latency devices.
- Fully handshake protocol for transfer reliability.
- Low power consumption.
- combines both **schedule-based as well as contention-based schemes.**
- The **protocol is asymmetric** in that different types of nodes with different roles are used.

IEEE 802.15.4 Applications Space



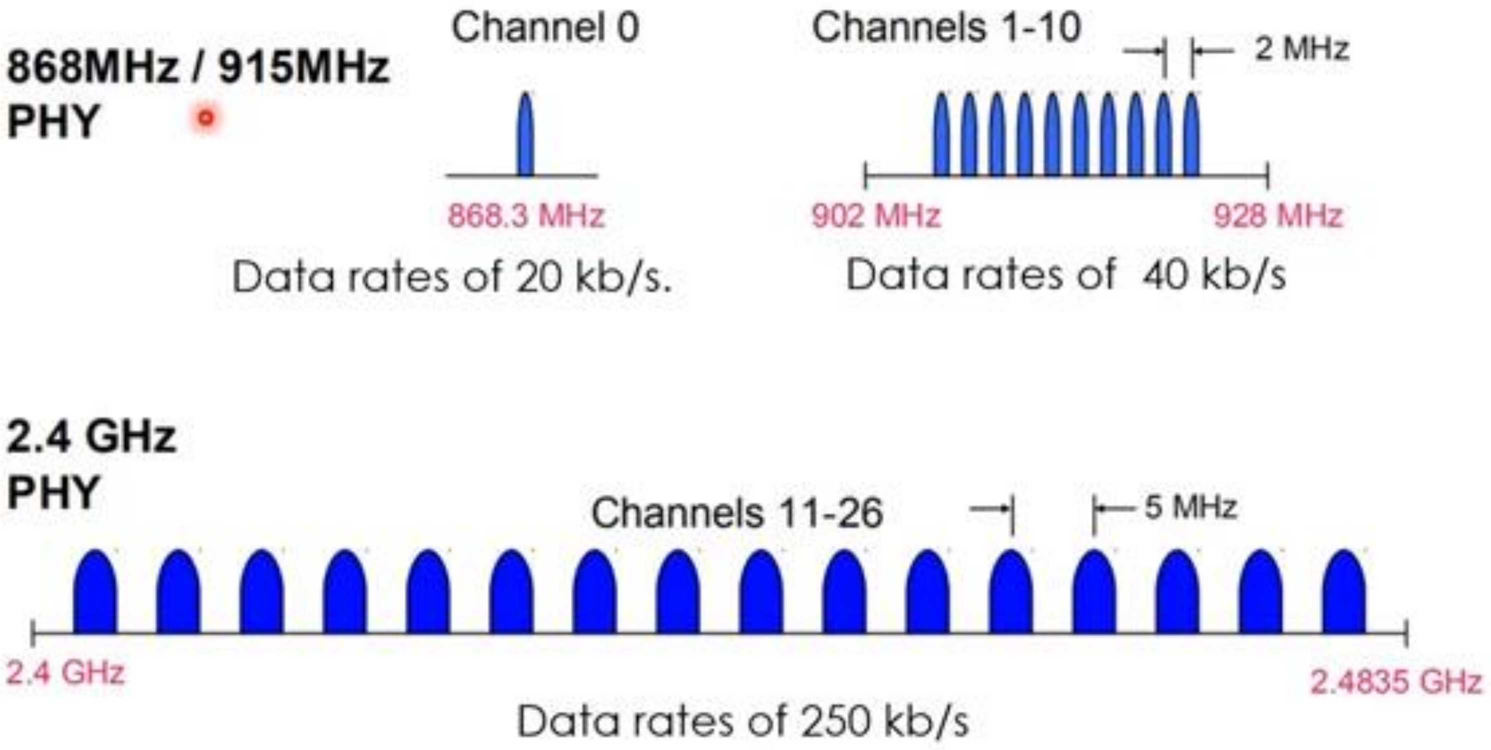
- Home Networking
- Automotive Networks
- Industrial Networks
- Interactive Toys
- Remote Metering



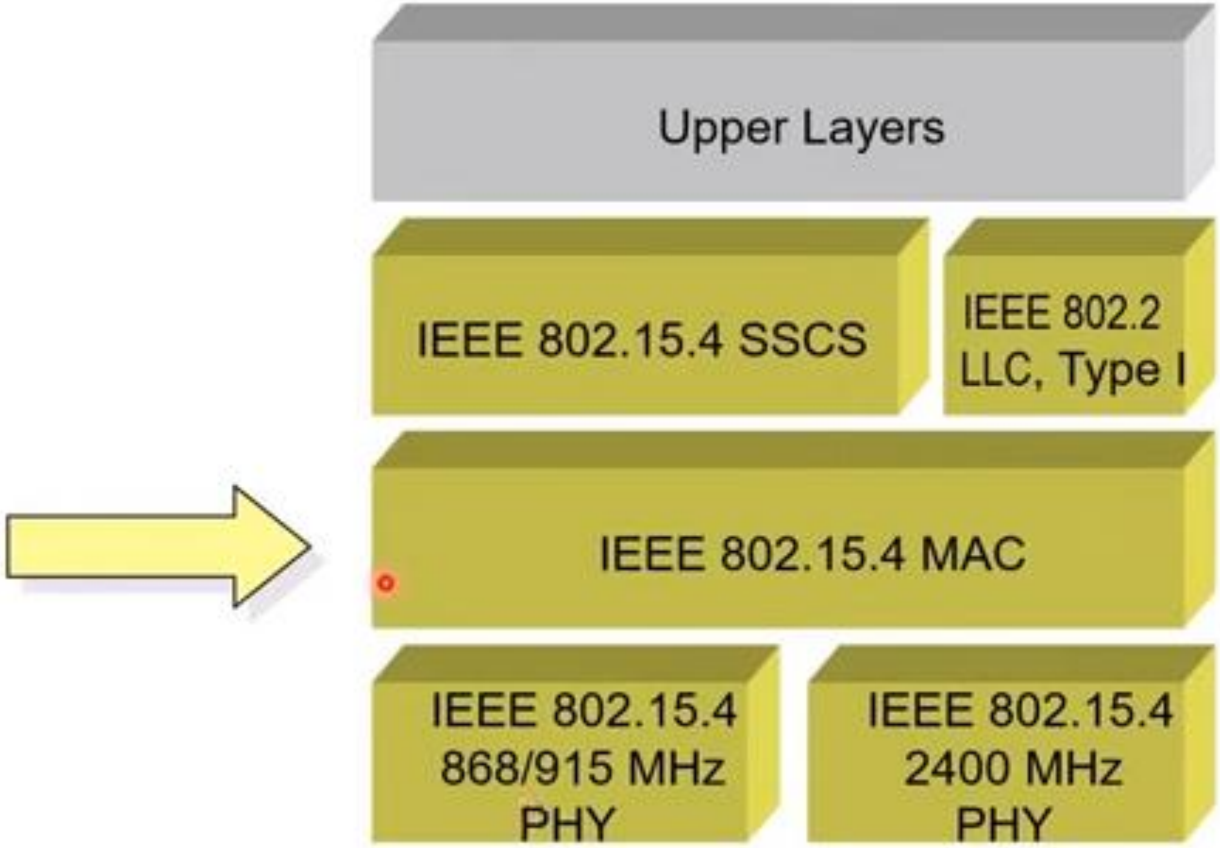
The targeted applications for IEEE 802.15.4 are in the area of wireless sensor networks, **home automation, home networking, connecting devices to a PC, home security, and so on.**

IEEE 802.15.4 PHY Overview

Operating Frequency Bands



802.15.4 Architecture



Network Architecture and Types & Roles of Nodes

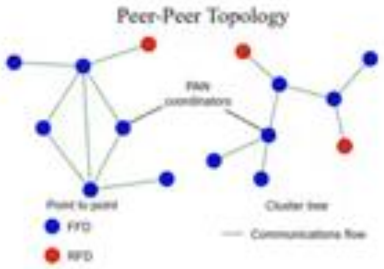
Two types of nodes:

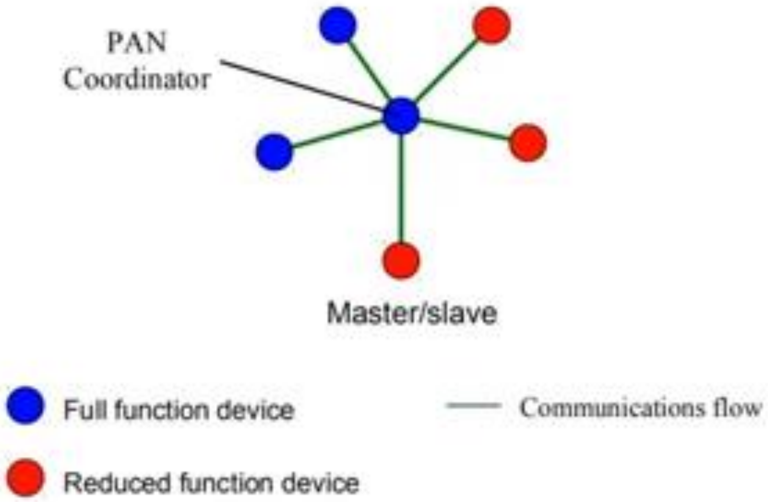
Full Function Device (FFD) can operate in three different roles:

1. PAN coordinator(PAN = Personal Area Network),
2. A simple coordinator
3. A device.

Reduced Function Device (RFD)

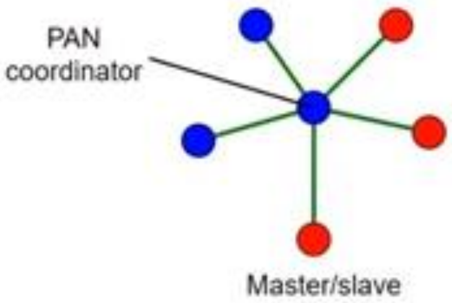
- operate only as a device.

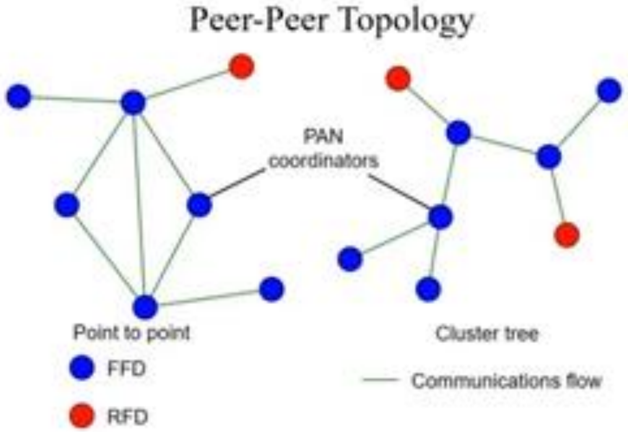




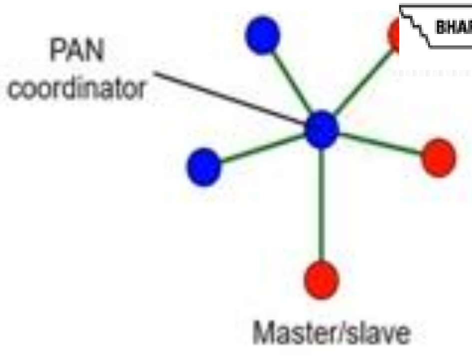
Cont.

- A device node must be associated to a coordinator node and communicates directly to the coordinator. **(forming a star network.)**
- Coordinators can operate in a **peer-to-peer fashion**
- **Multiple coordinators** can form a Personal Area Network (PAN).





A Coordinator Handles Following Tasks:

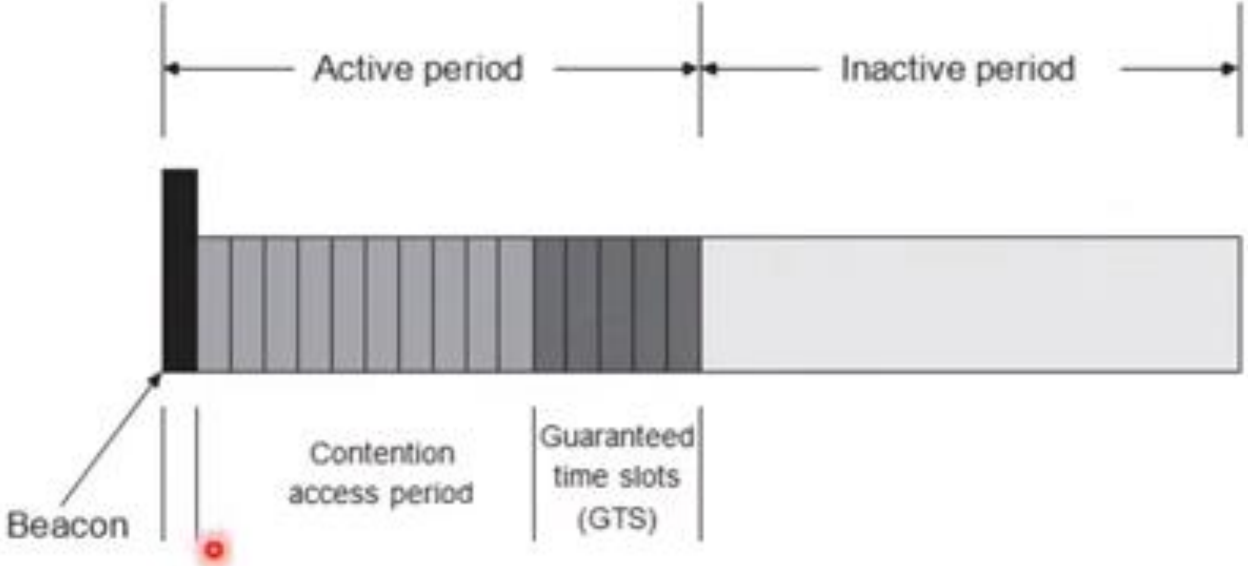


- It manages a list of associated devices.
- Devices are required to explicitly associate and disassociate with a coordinator using certain signaling packets.
- It allocates short addresses to its devices.
- All IEEE 802.15.4 nodes have a 64-bit device address.
- When a device associates with a coordinator, it may request assignment of a 16-bit short address to be used subsequently in all communications between device and coordinator.
- In the beaconed mode of IEEE 802.15.4, it transmits regularly frame beacon packets announcing the PAN identifier, a list of outstanding frames, and other parameters.
- Furthermore, the coordinator can accept and process requests to reserve fixed time slots to nodes and the allocations are indicated in the beacon.

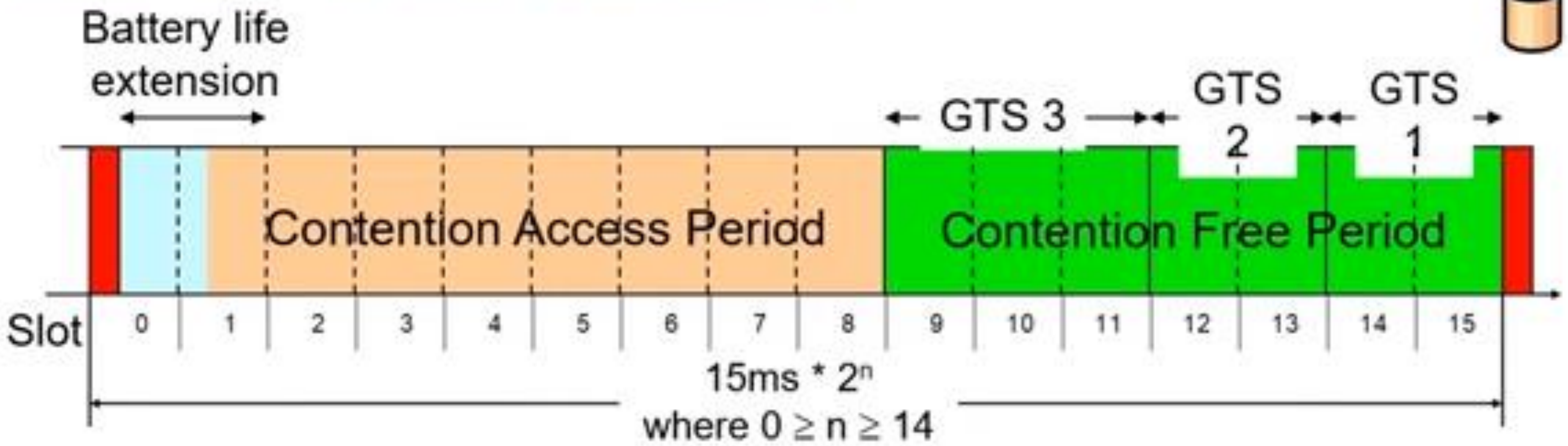
•It exchanges data packets with devices and with peer coordinators.

Superframe Structure

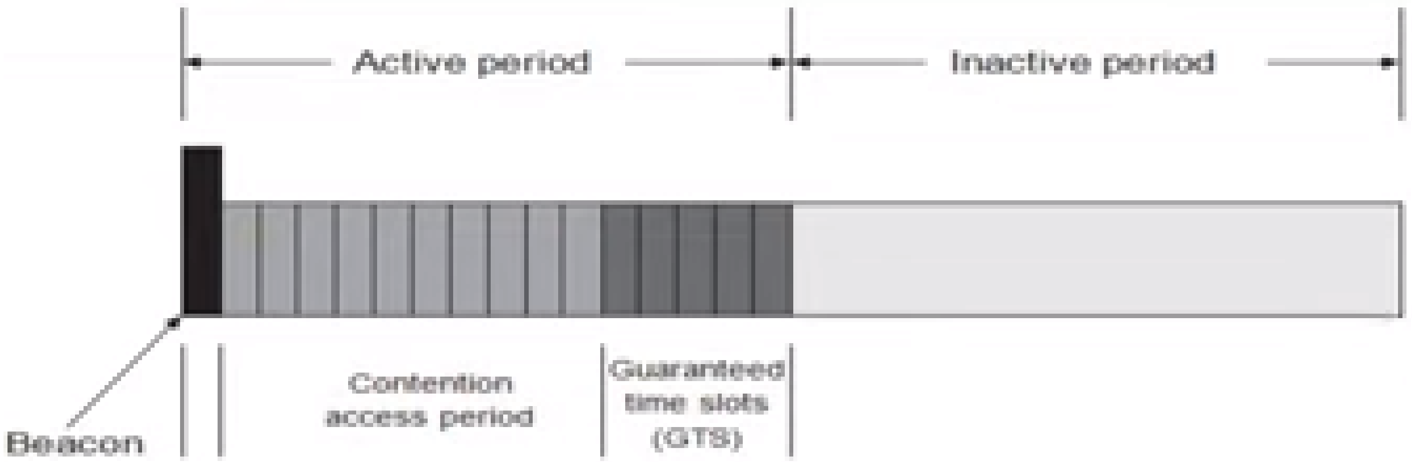
- We start with the beaconed mode of IEEE 802.15.4.



Optional Frame Structure

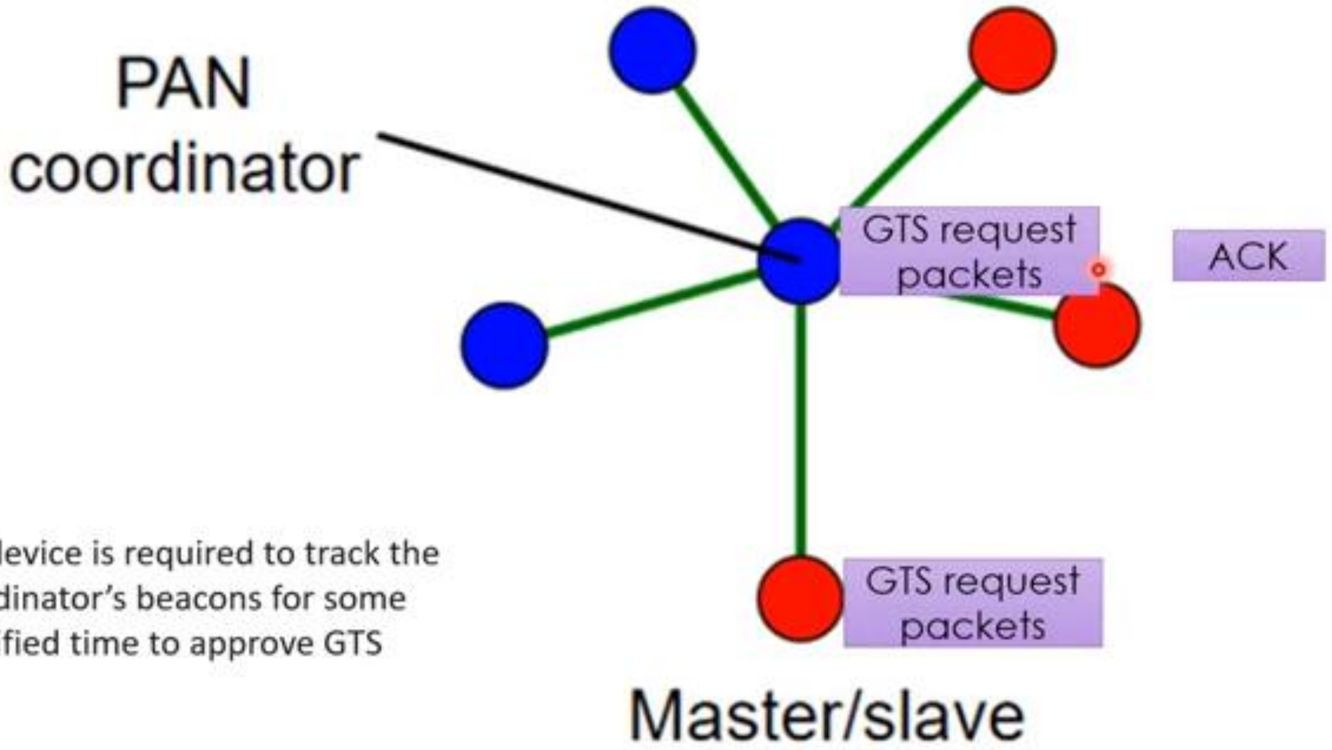


Network beacon	Transmitted by PAN coordinator. Contains network information, frame structure and notification of pending node messages.
Beacon extension period	Space reserved for beacon growth due to pending node messages
Contention period	Access by any node using CSMA-CA
Guaranteed Time Slot	Reserved for nodes requiring guaranteed bandwidth.



- The length of the active and inactive period as well as the length of a single time slot and the usage of GTS slots are configurable.
- The coordinator is active during the entire active period.
- The **associated devices are active in the GTS phase only in time slots allocated to them**; in all other GTS slots they can enter sleep mode.
- In the **Contention Access Period** (CAP), **a device node can shutdown its transceiver.**
- Coordinators do much more work than devices and the protocol **is inherently asymmetric.**

Guaranteed Time Slots (GTS) Management

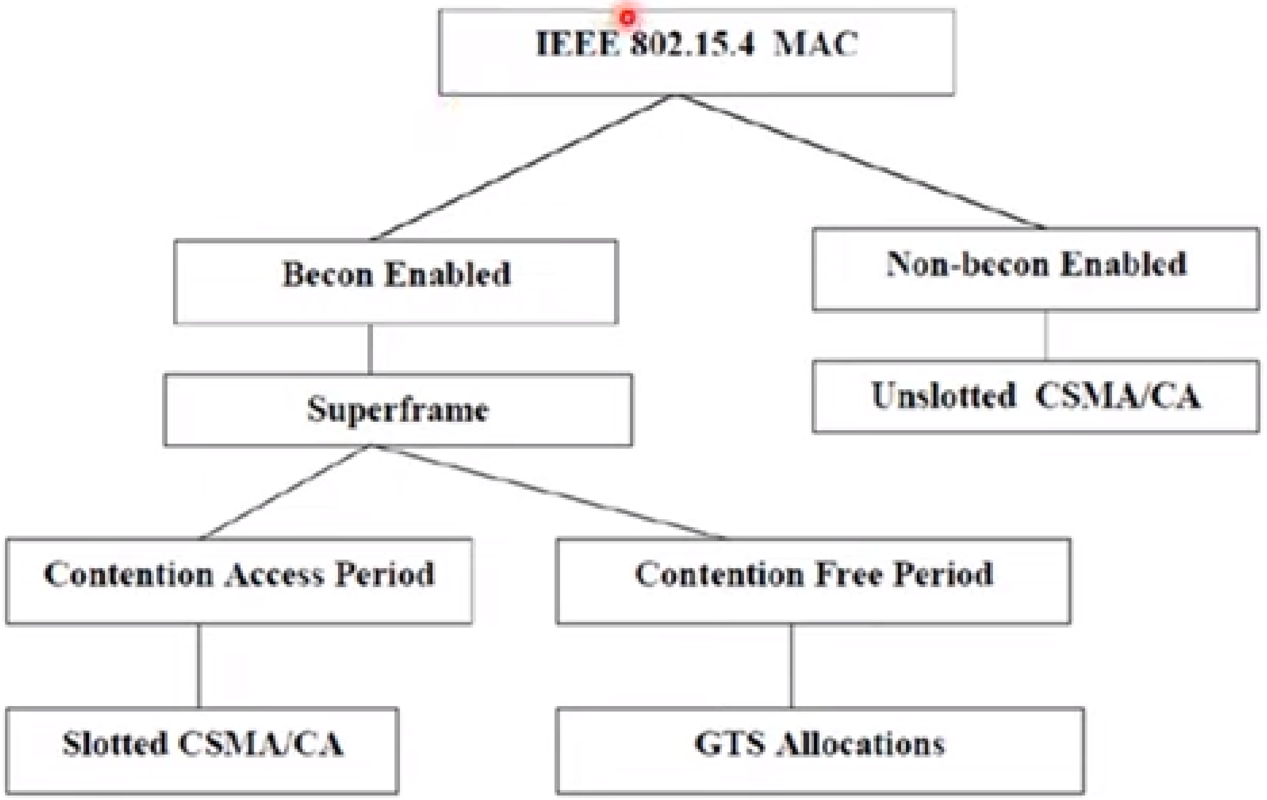


the device is required to track the coordinator's beacons for some specified time to approve GTS

Cont.

- If the coordinator has insufficient resources, it generates a GTS descriptor for (invalid) time slot zero, indicating the available resources in the descriptors length field.
- the device may consider renegotiation.
- it concludes that the allocation request has failed.
- A GTS is allocated to a device on a regular basis until it is explicitly deallocated.
- The deallocation can be requested by the device by means of a special control frame.
- After sending this frame, the device shall not use the allocated slots any further.
- The coordinator can also trigger deallocation based on certain criteria. Specifically, the coordinator monitors the usage of the time slot: If the slot is not used at least once within a certain number of superframes, the slot is deallocated.
- The coordinator signals deallocation to the device by generating a GTS descriptor with start slot zero.

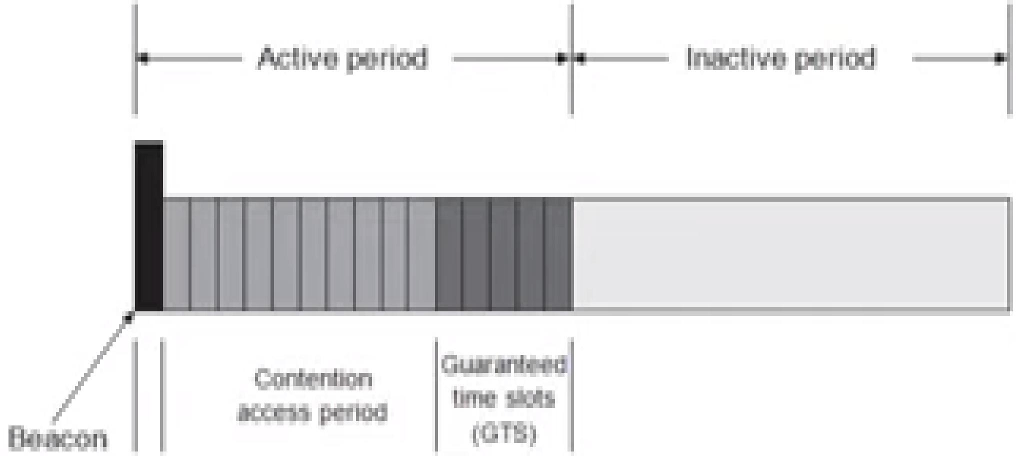
Operational Models Of IEEE 802.15.4



Data Transfer Procedures

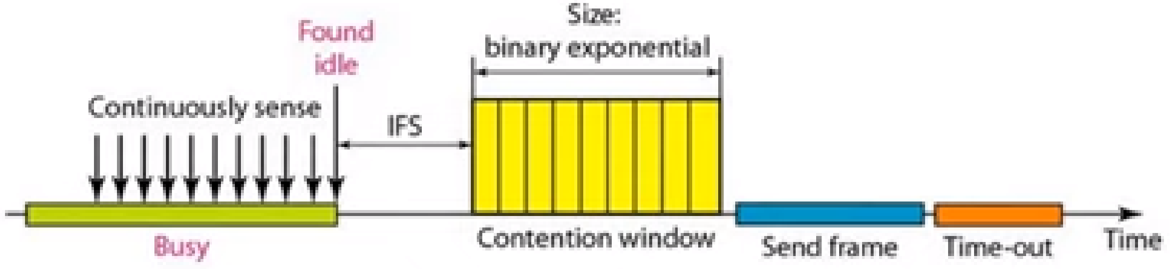
- first assume that a device wants to transmit a data packet to the coordinator

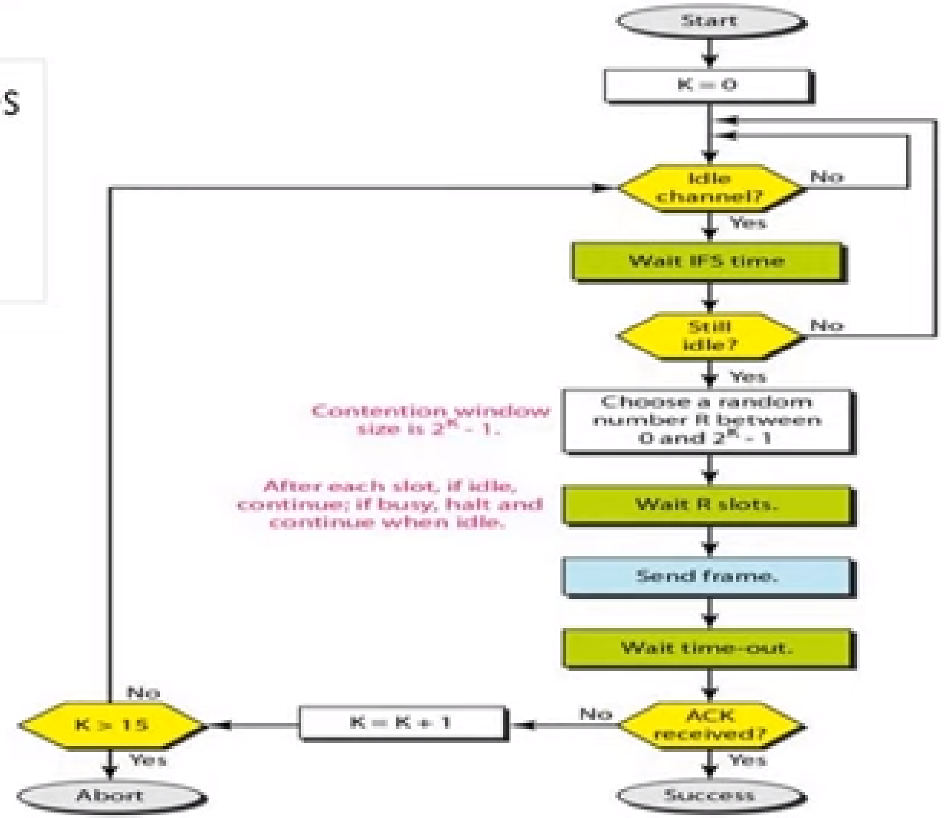
If the device has allocated a receive GTS and when the packet/acknowledgment/IFS cycle fits into these, the coordinator simply transmits the packet in the allocated time slot without further coordination.



Slotted CSMA-CA protocol

Collision are avoided using three CSMA/CA strategies
- Inter Frame Space
- Contention Window
- Acknowledgements





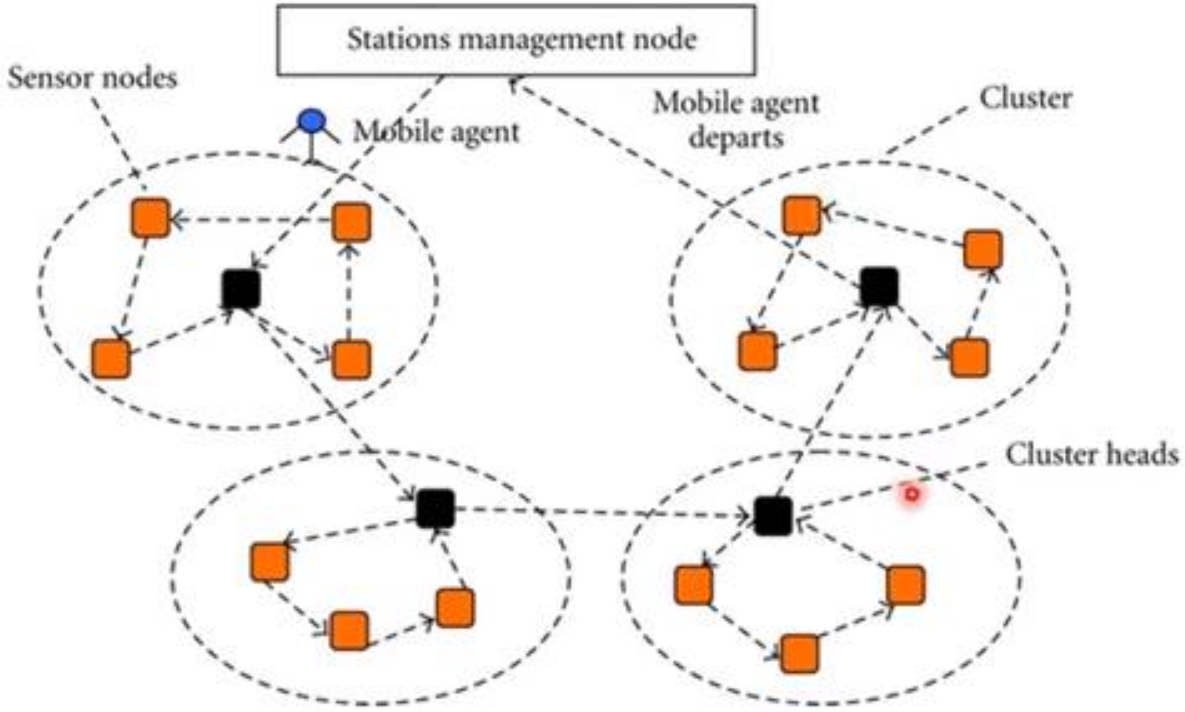
Nonbeaconed mode/ unslotted CSMA-CA protocol

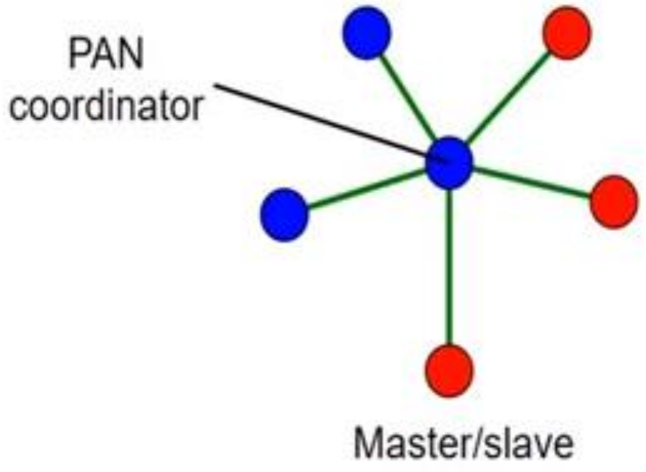
- In the nonbeaconed mode, the coordinator **does not send beacon frames nor is there any GTS mechanism.**
- The lack of beacon packets takes away a good opportunity for devices to acquire **time synchronization with the coordinator.**
- All packets from devices are transmitted using an unslotted CSMA-CA protocol.
- the device performs only a single CCA (Clear Channel Assessment) operation. If this indicates an idle channel, the device infers success.
- **Coordinators must be switched on constantly** but devices can follow their own sleep schedule.
- Devices wake up for two reasons:

(i) to send a data or control packet to the coordinators,

(ii) to fetch a packet from the coordinator

Objective of Routing protocols





Routing protocols

- In a multihop network, **intermediate nodes have to relay packets from the source to the destination node.**
- Such an intermediate node **has to decide to which neighbor to forward an incoming packet.**
- Typically, **routing tables that list the most appropriate neighbor for any given packet destination are used.**
- The **construction and maintenance of these routing tables is the crucial task** of a distributed routing protocol.

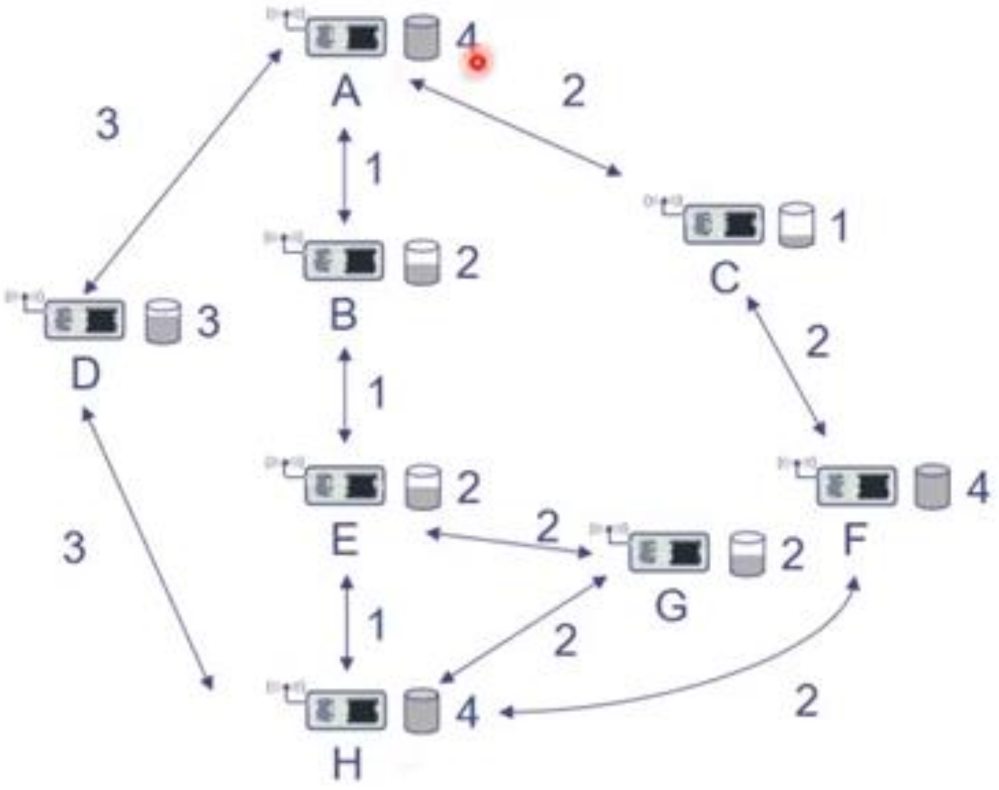
Energy-efficient unicast Routing

1. Minimize energy per packet (or per bit)
2. Maximize network lifetime
3. Routing considering available battery energy
    1. Maximum Total Available Battery Capacity
    2. Minimum Battery Cost Routing (MBCR)
    3. Min– Max Battery Cost Routing (MMBCR)
    4. Conditional Max – Min Battery Capacity Routing (CMMBCR)
    5. Minimize variance in power levels
    6. Minimum Total Transmission Power Routing (MTPR)

Energy-efficient unicast

- At a first glance, **energy-efficient unicast routing** appears to be a simple problem:
- **Take the network graph, assign to each link a cost value that reflects the energy consumption across this link, and pick any algorithm that computes least-cost paths in a graph.**
- shortest path algorithm to obtain routes with **minimal total transmission power.**
- In fact, there are various aspects how energy or power efficiency can be considered of in a routing context.
- Following Figure shows an example scenario for a **communication between nodes A and H** including link energy costs and available battery capacity per node.

Example scenario for a communication between nodes A and H

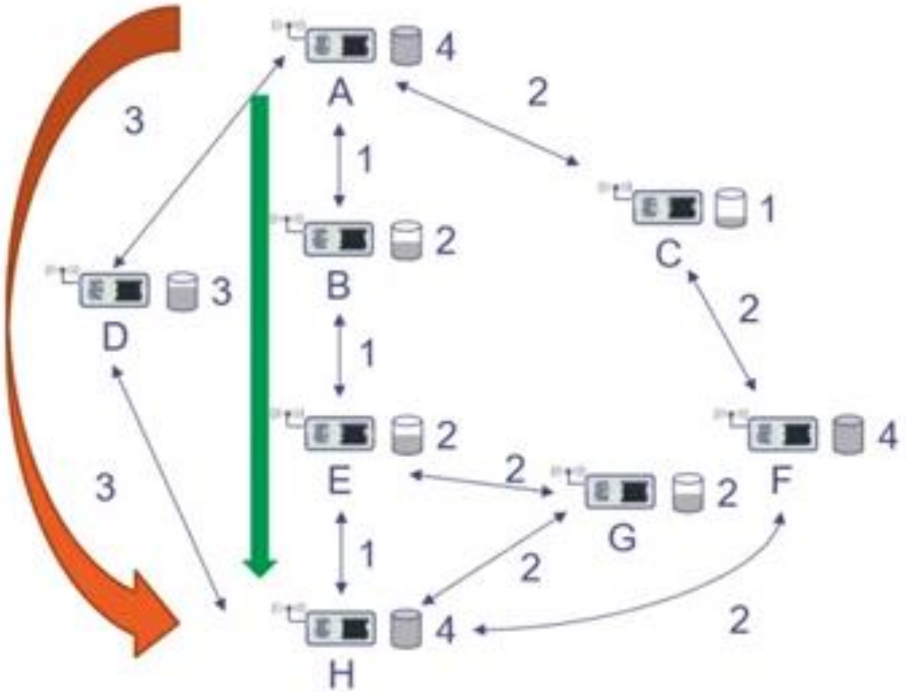


Example: Send data from node A to node H

Path-1= ACFH
Path-2= ABEGH
Path-3=ABEH
Path-4 =ADH

Minimize Energy Per Packet (or per bit)

- The most straightforward formulation is to look at the total energy required to transport a packet over a multihop path from source to destination..
- The goal is then to minimize, for each packet, by selecting a good route.
- Minimizing the hop count will typically not achieve this goal as routes with few hops might include hops with large transmission power to cover large distances
- However, this cost metric can be easily included in standard routing algorithms.
- <u>Based on cost of the link per path</u>
- Path-1= ACFH=2+2+2=6 Units of power
- Path-2= ABEGH= 1+1+2+2=4 Units of power
- Path -3=ABEH=1+1+1=3 Units of power
- Path-4 =ADH=3+3=6 Units of power



Example: Send data from node A to node H

Maximize network lifetime

- A WSN's task is not to transport data, but to observe (and possibly control).
- Hence, energy-efficient transmission is at best a means to an end

- **Which event to use to differentiate the end of a network's lifetime is,**

1. Time until the first node fails.
2. Time until there is a spot that is not covered by the network
3. Time until network partition (when there are two nodes that can no longer communicate with each other

Maximum Total Available Battery Capacity

Choose that route where the sum of the available battery capacity is maximized

Based on available battery units per path

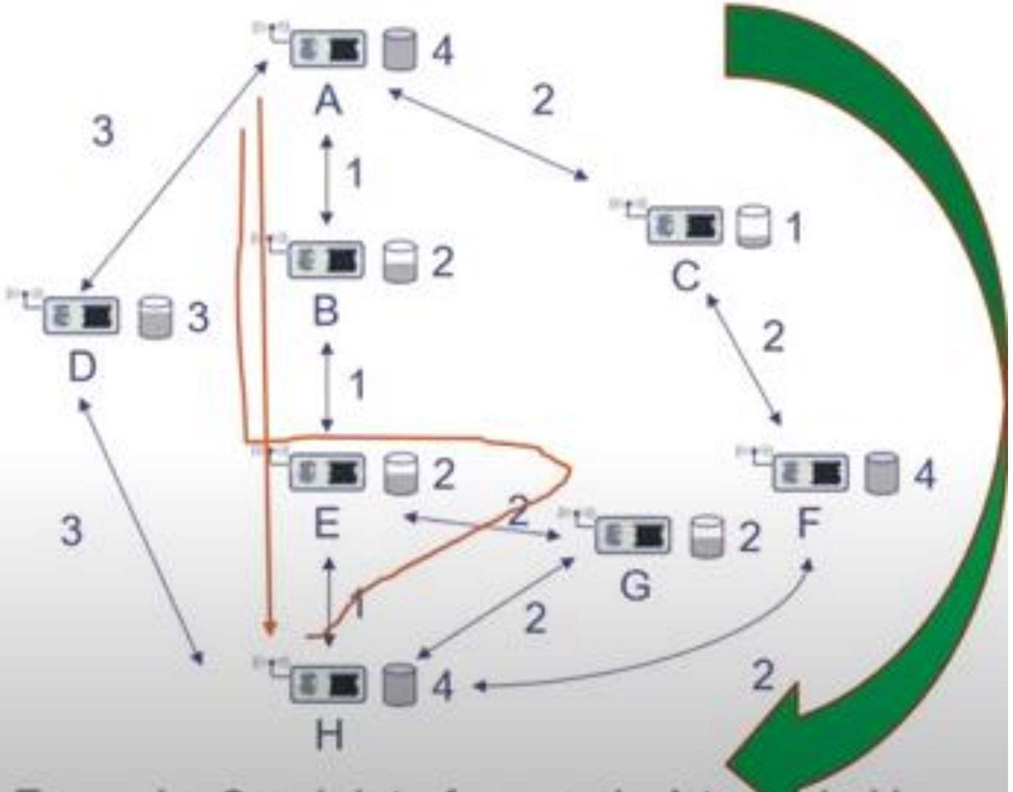
Path-1= ACFH=1+4 =5 Units of power – Selected

Path-2= ABEGH= 2+2+2 =6 Units of power- Not Selected

Path -3=ABEH=2+2 =4 Units of power

Path-4 =ADH=3 =3 Units of power

Looking only at the intermediate nodes route **A-B-E-G-H has a total available capacity of 6 units,** but that is only because of the extra node G that is not really needed – such detours can of course arbitrarily increase this metric. **Hence, A- B-E-G-H should be discarded as it contains A-B-E-H as a proper subset**.



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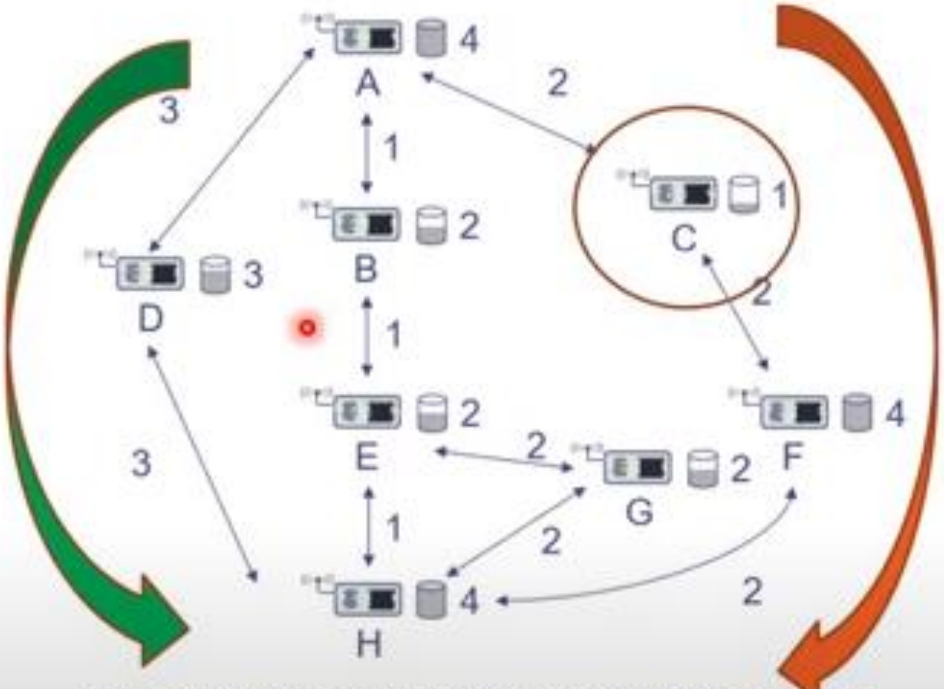
Minimum Battery Cost Routing (MBCR)

- Instead of looking directly at the sum of available battery capacities along a given path,
- MBCR looks at the "reluctance" of a node to route traffic.
- This reluctance increases as its battery is drained;
- for example, **routing cost can be measured as the reciprocal of the battery capacity**.
- Then, the cost of a path is the sum of this reciprocals and **<u>the rule is to pick that path with the smallest cost</u>**.

- **Path-1= ACFH= 1/1 + 1/4 = 1.25**
- **Path-2= ABEGH= ½+½+½ =1.5**
- **Path -3=ABEH= 1/2+ 1/2 = 1**
- **Path-4 =ADH= 1/3 =0.33- Selected**

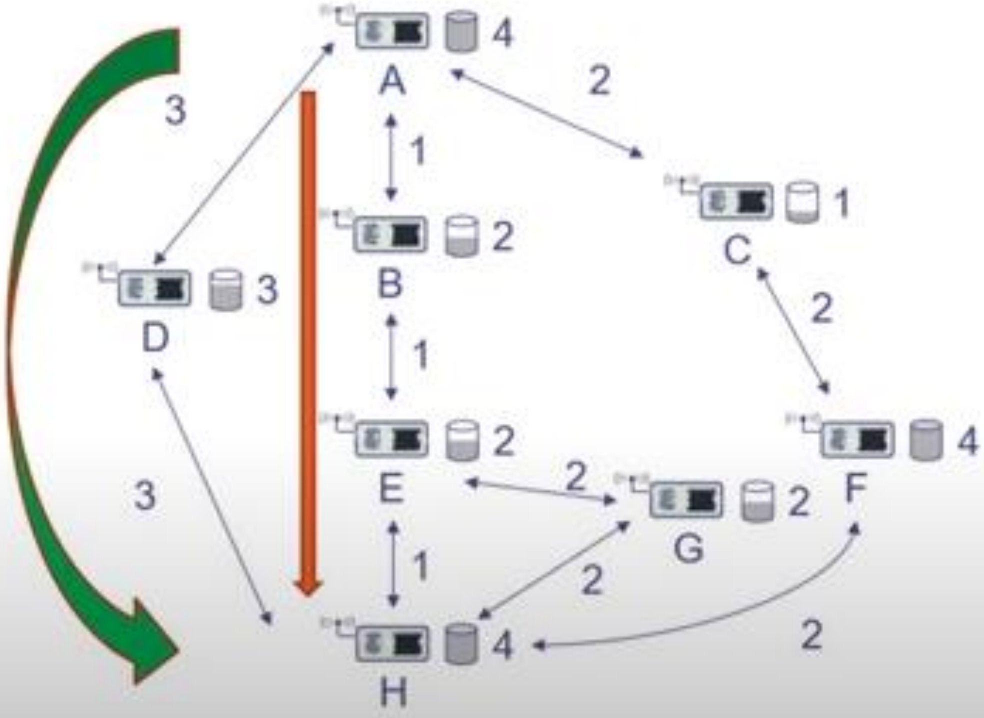
- **Since the reciprocal function assigns high costs to**



Example: Send data from node A to node H

Conditional Max – Min Battery Capacity Routing (CMMBCR)

- **Minimize Energy Per Packet**
- **Minimum Battery Cost Routing (MBCR)**
- Another option is to conditionalize upon the actual battery power levels available. If there are routes along which **all nodes have a battery level exceeding a given threshold, <u>then select the route that requires the lowest energy per bit.</u>**
- **If there is no such route, then pick that route which maximizes the minimum battery level. (<u>Minimum Battery Cost Routing</u>)**



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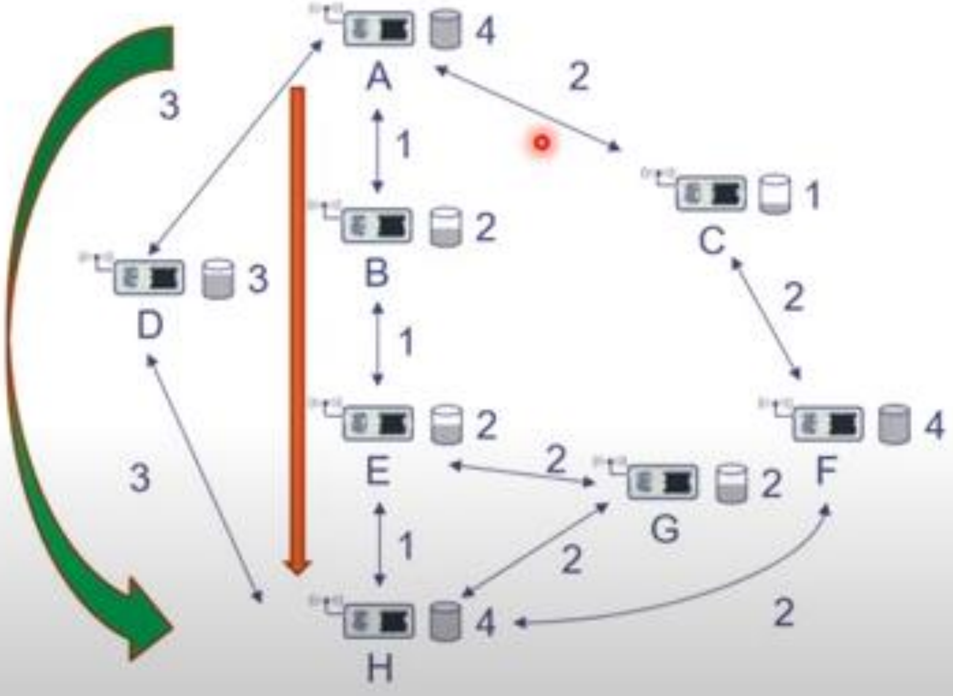


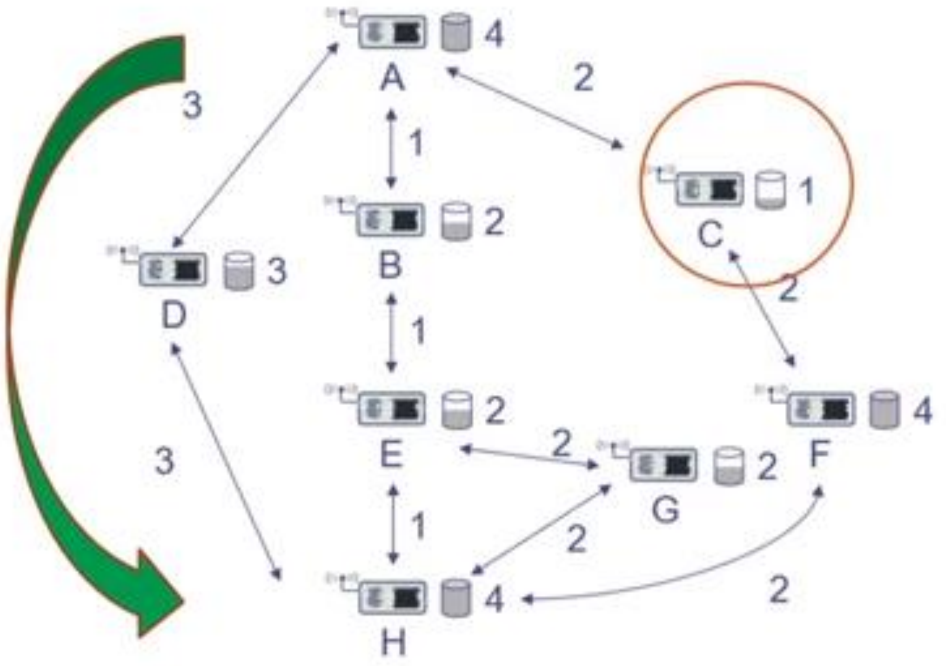
Example: Send data from node A to node H

Minimize Variance in power levels

- **To ensure a long network lifetime,** one strategy is to use up all the batteries uniformly to avoid some nodes prematurely running out of energy and disrupting the network.
- **Hence, routes should be chosen such that the variance in battery levels between different routes is reduced.**

Path-1= ACFH
Path-2= ABEGH
Path-3=ABEH
Path-4 =ADH

A=4	E=2
B=2	F=4
C=1	G=2
D=3	H=4



Example: Send data from node A to node H

Minimum Total Transmission Power Routing (MTPR)

- looked at the situation of **several nodes transmitting directly to their destination, mutually causing interference with each other.**
- A given transmission is successful if its SINR (signal-to-interference-plus-noise ratio) exceeds a given threshold.
- The goal is to find an assignment of transmission power values for each transmitter (given the channel attenuation metric) such that all transmissions are successful and that the sum of all power values is minimized.
- MTPR is of course also applicable to multihop networks.
- **A direct performance comparison between these concepts is difficult as they are trying to fulfill different objectives.**

Geographical Routing

- It makes use of location information in routing
- The location of both the sender and the destination is known.

Wireless sensor networks (**WSNs**) refer to networks of spatially dispersed and dedicated sensors that monitor and record the physical conditions of the environment and forward the collected data to a central location.

Geographic routing (also called **georouting** or **position-based routing**) is a routing principle that relies on geographic position information. It is mainly proposed for wireless networks and based on the idea that the source sends a message to the geographic location of the destination instead of using the network address.

Introduction

Routing in wireless sensor networks is performed in a cooperative way, i.e. each node relies on its neighbours to forward its packets towards the network sink.

All sensor nodes participate in the routing procedure acting as routers. This intricacy of the sensor networks implies that the routing functionality is distributed over all network nodes, which however have limited energy, memory and processing capabilities.

Cont.

Although a plethora of routing protocols for ad hoc and sensor networks has been proposed, geographical routing (GR) targets the efficient support of mobility and scalability while it turns out to be invulnerable against a set of security attacks when security comes into play.

GR outperforms other solutions because it addresses at the same time scalability and mobility.

The concept is to use geography for routing instead of measuring hops and flooding the current state of all network nodes to create a map.

Cont.

The idea is that each node is characterized by its coordinates and each time a node needs to send a packet, it will forward it to the neighboring node that is closer to the destination than all the rest nodes.

To perform this decision, it is required to keep geographical information (the location) for each neighbour in its routing table.

This information is gathered through the so-called **“BEACON”** messages that each node periodically transmits announcing its location.

Cont.

This localized operation has the following important advantages:

- It facilitates the mobility support.
- It is scalable.
- It introduces minimal routing overhead.

GR is a proactive routing protocol since beacons are emitted periodically. The disadvantage of proactive routing protocols is that routing messages are exchanged even when no traffic to route exists.

Cont.

The realization of GR algorithms requires that each node is aware of its coordinates either due to the existence of a GPS (Global Positioning System) device in each sensor node or based on the implementation of some localization functionality.

This aspect has attracted much attention since it directly affects the cost of the sensor node and the achieved security

GEOGRAPHICAL ROUTING PROTOCOLS

- ❖Greedy Perimeter Stateless Routing (GPSR)
- ❖Geographical and Energy Aware Routing (GEAR)
- ❖Energy Aware Greedy Routing (EAGR)
- ❖Probabilistic Geographic Routing (PGR)
- ❖Energy-aware Geographical Forwarding using Adaptive Sleeping (EnGFAS)
- ❖Directional Location-based Randomized Routing (DLR)
- ❖On Demand GPSR (OD-GPSR)
- ❖Blind Geographic Routing (BGR)